

1. Biochemistry

Power to Understand What We Are!

Course information

- **Teaching Methods:**
- Lecture
- Demonstration
- Presentation and group discussion
- **Assessment Method:**
- Assignment & Quiz (30%)
- Mid examination (30%)
- Final examination (40%)
- **Time Allocation:** 48 contact hours
- **Grading:** Fixed scale

Course objectives

At the end of the course the students should be able to:

- Describe the classification, properties and function of the four major biomolecules
- Gain basic insight into the dynamic events of metabolism, so characteristic of every living cell
- Identify the common tests used for carbohydrates and protein
- Discuss the role of enzymology in pharmacology and clinical diagnosis
- Understand the storage and expression of genetic information
- Discuss vitamins and minerals

Course outline

1.0. Introduction

1.1. Definition of biochemistry

1.2. The scope of biochemistry

1.3. Significance of biochemistry in medicine

1.4. The cell and major chemical constituents of cells
(water, acid and base)

1.5. Water as a solvent and as a medium for
biochemical reactions

Outline cont'd

1.6. Chemical properties and structure of water

1.7. Water and thermal regulation

1.8. Electrolyte

1.9. Osmolality and water movement

2.0. Acid and base

Outline cont'd

2.0. Carbohydrates

- 2.1. Introduction and classification-mono,di- and polysaccharides
- 2.2. Chemistry and properties and their test
- 2.3. Carbohydrate metabolism
- 2.4. Hormonal regulations of carbohydrate metabolism
- 2.5. Clinical significances in carbohydrate metabolism

Outline cont'd

2.6. Glycolysis

2.7. Rapaport luebering shunt

2.8. Gluconeogenesis

2.9. Glycogen metabolism

2.10. Pentose monophosphate pathway

2.11. TCA cycle

Outline cont'd

3.0. Proteins

- 3.1. Classification, naming, Structure and function of amino acids
- 3.2. Structure of proteins (conformation)
- 3.3. Functions of proteins
- 3.4. Myoglobin and Hemoglobin Structure

Outline cont'd

- 3.5. Protein metabolism
- 3.6. Essential and non-essential aminoacids
- 3.7. Catabolism aminoacids
- 3.8. Transamination and oxidative deamination
- 3.9. Urea cycle
- 3.10. Catabolism of aminoacid carbon skeleton
- 3.11. Inborn errors of aminoacid metabolism
- 3.12. Heme Metabolism

Outline cont'd

4.0. Enzymes

4.1. Nomenclature

4.2. Properties of enzymes

4.3. Theory of enzyme catalysis

4.4. Factors affecting the velocity of enzyme-catalyzed reactions

4.5. Enzyme inhibitors

4.6. Enzymes in clinical diagnosis and pharmacology

Outline cont'd

5.0. Vitamins

5.1. Classification-water soluble and fat soluble

5.2. Sources, physiological functions and deficiency of water soluble Vitamins- B vitamins and vitamin C

5.3. Sources, physiological functions and deficiency of fat soluble vitamins- vitamin A, vitamin D, vitamin K and vitamin E

Outline cont'd

6. Lipids

6.1. Classification

6.2. Structure and properties of fatty acids, phospholipids, triacylglycerol, Cholesterol, lipoproteins and ketone bodies

6.3. Lipid metabolism

Outline cont'd

6.4. Lipid transport

6.5. Lipoproteins

6.6. Clinical correlates

6.7. Ketosis

6.8. Disorders of plasma lipoprotein
(hypercholesterolemia and atherosclerosis)

outline cont'd

7.0. Oxidative phosphorylation and electron transport chain Oxidation and Reduction

7.1. The Electron Transport Chain

7.2. Connections and Regulation

7.3. Ratios of phosphate to oxygen

7.4. Uncouplers

7.5. Inhibitors

Outline cont'd

8.0. Nucleic acids

8.1. Classification and chemistry of nucleic acids-RNA and DNA

8.2. Metabolism of nucleotides

8.3. Clinical significance (Hyperuricemia and gout and Orotic aciduria)

8.4. DNA replication

8.5. Transcription

8.6. Translation

1.0. Introduction

Learning objectives

At the end of this session the students will be able to:-

- Explain the definition ,the scope and significance of biochemistry .
- Describe the major chemical constituents of cells.
- Discuss the role of organelles.

What is Biochemistry?

- Simplest definition: “Chemistry of the living cell”.
- The science concerned with **the chemical basis of life**.
- Like: **chemical constituents, reactions and other processes which undergo in living cells** .
- Biochemistry is the study of the **structure**, **composition**, and **chemical reactions** of substances in **living systems**.

...cont'd

- Generally biochemistry is **the study of life at the molecular level.**
- Uses basic laws of chemistry, biology and physics to explain processes of living cells.
- Because life depends on **biochemical reactions**, biochemistry has become the **basic language of all life sciences.**

Why study biochemistry?

- **Lead us to fundamental understanding of life.**
- **Understand important issues in medicine, health, and nutrition.**
 - ☛ Has led to greater molecular understanding of diseases such as diabetes, sickle cell anemia, and cystic fibrosis.
 - ☛ Next frontier: AIDS, cancer, Alzheimer's Disease.
- **Advance biotechnology industries**
 - ☛ Biotechnology is the application of biological cells, cell components, and biological properties to technically and industrially useful operations.

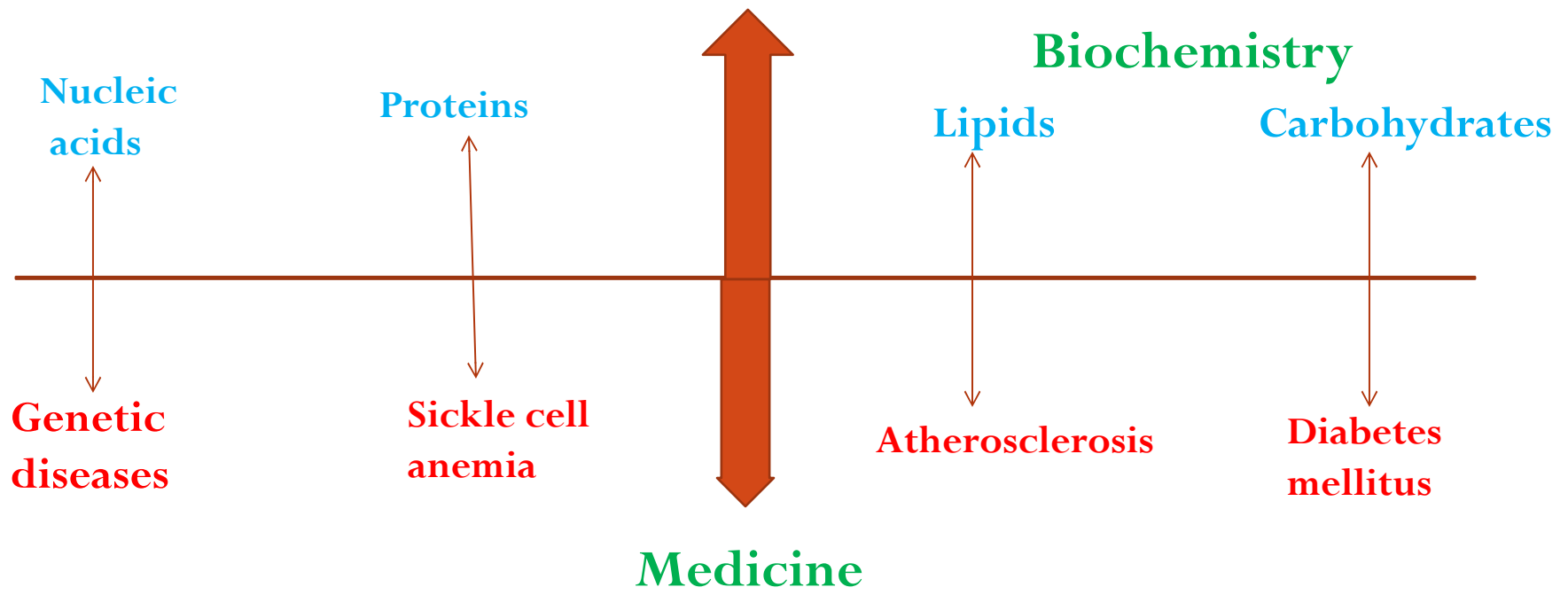
i. Major objective of biochemistry

- Complete understanding at the **molecular level** of all the chemical processes associated with living cells.
 - **Isolate**-the numerous molecules found in cells,
 - **Determine**-their structures,
 - **Analyze**-how they function.

ii. Further objective

- Attempt to understand how life began.
- An appreciation of the biochemistry of less complex form of life is often direct relevance to human biochemistry.

Fig.1 Examples of the two-way street connecting biochemistry and medicine.



1.2 The major chemical constituents of cells

Biochemistry = the chemistry of life

Elements - These are single substances which cannot be broken down any more. there are 110 different elements that are known to man.

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89 Ac	104 Unq	105 Unp	106 Unh	107 Uns	108 Uno	109 Une	110 Unn								

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

- The living matter is composed of mainly *six elements*- **C**arbon, **H**ydrogen, **O**xygen, **N**itrogen, **P**hosphorus and **S**ulphur. (CHONPS)
- These elements together constitute about *90% of the dry weight of the human body*.
- Several other functionally important elements are also found in the cells.
- These include Ca, K, Na, Cl, Mg, Fe, Cu, Co, Zn, F, Mo and Se.

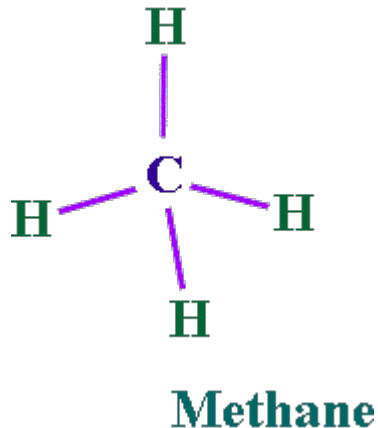
Carbon-a unique element of life

- Carbon is the *most predominant and versatile element* of life.
- It possesses as *unique property* to form infinite number of compounds.
- This is attributed to the ability of carbon to form:
 - **stable covalent bonds** and *C-C chains of unlimited length.*
- It is estimated that about 90% of compounds found in living system invariably contain carbon.

Carbon can form immensely diverse compounds, from simple to complex.

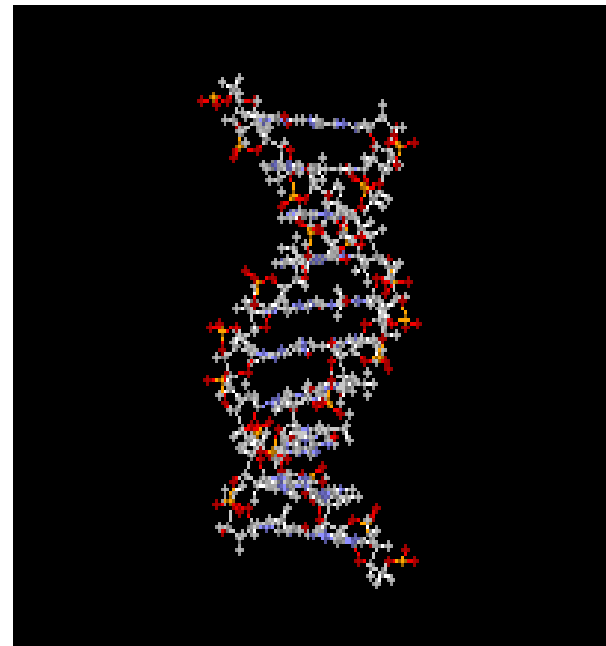
Compounds - These are two or more elements combined.
These elements are bonded together.

Simple Molecule



Methane with 1
Carbon atom

Complex Molecule



DNA with tens of
Billions of Carbon atoms

Organization of Life

- **Life** is composed of *lifeless chemical molecules*.
- elements
- simple organic compounds (monomers)
- macromolecules (polymers)
- supramolecular structures
- organelles
- cells
- tissues
- organisms

Complex Biomolecules

- As regards **lipids**, it may be noted that *they are not biopolymers in a strict sense*, but majority of them contain fatty acids.

<i>Biomolecule</i>	<i>Building block (repeating unit)</i>	<i>Major functions</i>
1. Protein	Amino acids	Fundamental basis of structure and function of cell (static and dynamic functions).
2. Deoxyribonucleic acid (DNA)	Deoxyribonucleotides	Repository of hereditary information.
3. Ribonucleic acid (RNA)	Ribonucleotides	Essentially required for protein biosynthesis.
4. Polysaccharide (glycogen)	Monosaccharides (glucose)	Storage form of energy to meet short term demands.
5. Lipid	Fatty acids, glycerol	Storage form of energy to meet long term demands; structural components of membranes.

Composition of the body and major classes of molecules

- The human body is composed of a few **elements** that combine to form a great variety of **biomolecules**.
- Biomolecules are compounds of carbon with a variety of functional groups

Element

- The simplest kind of matter which can not be split in to two or more simpler substances by chemical reactions.

Chemical composition of a normal man (weight 65 kg)

Constituent	Percent (%)	Weight (kg)
Water	61.6	40
Protein	17.0	11
Lipid	13.8	9
Carbohydrate	1.5	1
Minerals	6.1	4

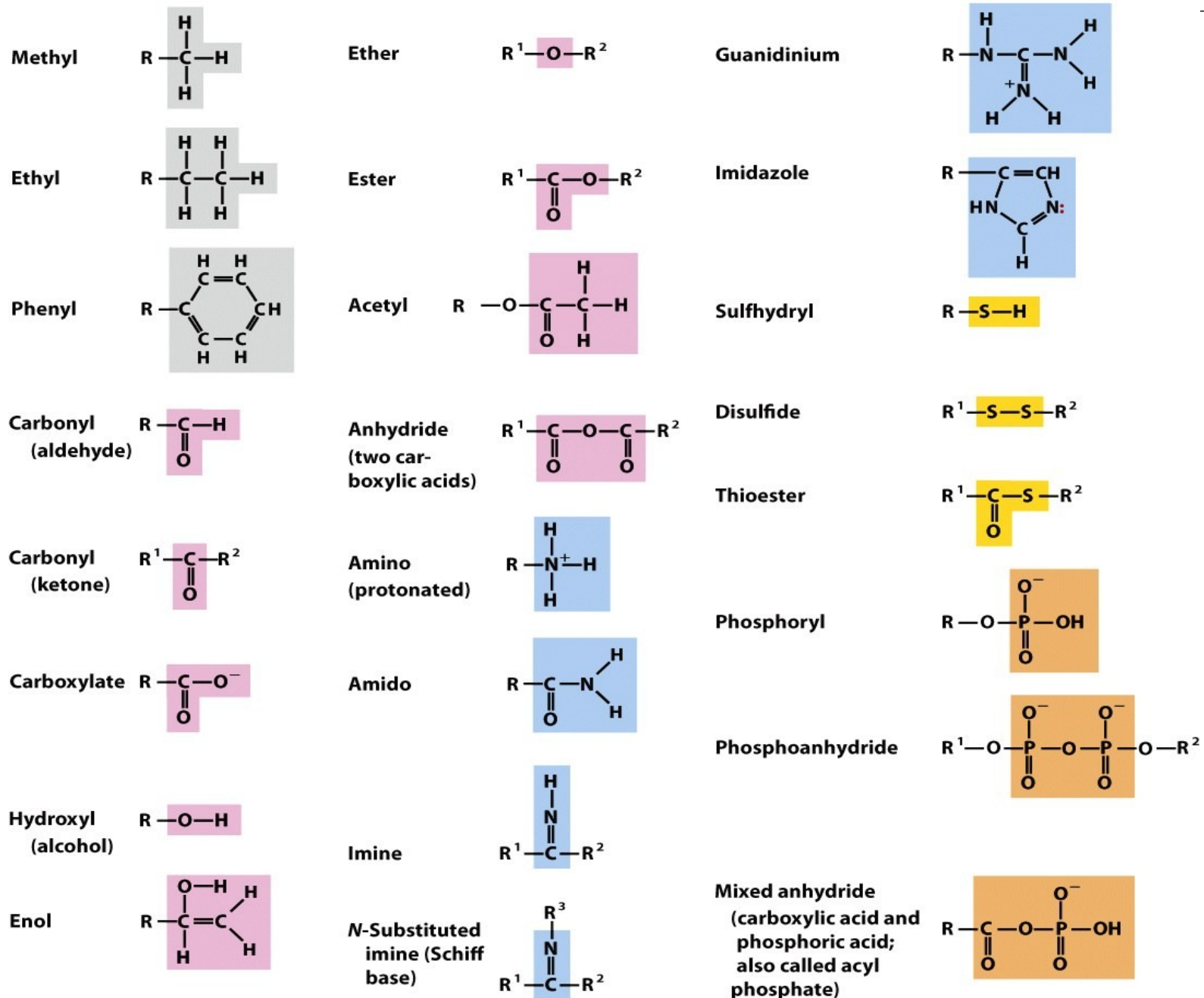


Fig. Some common functional groups of

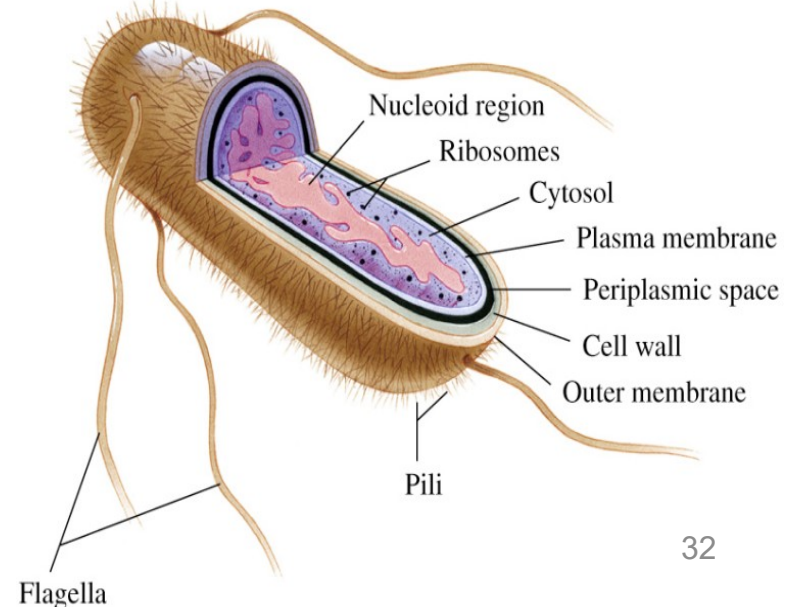
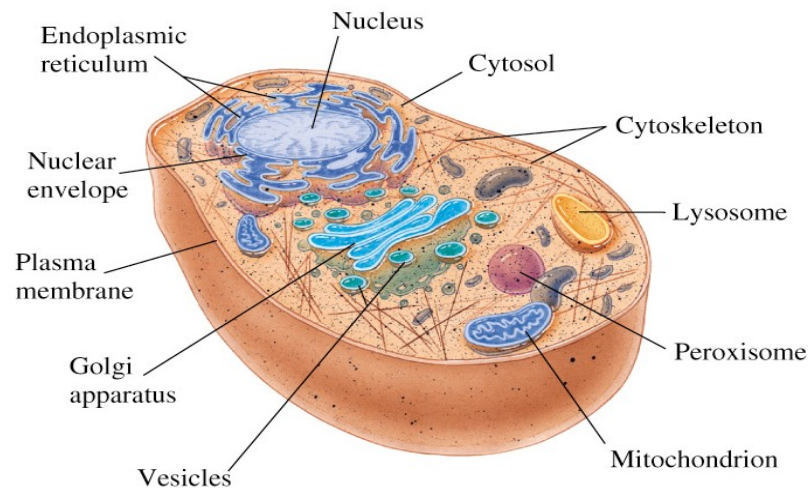
biomolecules

Cells

- Basic building blocks of life.
- Smallest living unit of an organism.
- Grow, reproduce, use energy, adapt, respond to their environment.
- Many cannot be seen with the naked eye.
- A cell may be an entire organism or it may be one of billions of cells that make up the organism.

Basis Types of Cells

(a)



Similarities among all types of cells

All cells use nucleic acids (**DNA**) **to store information**

Except RNA viruses, but not true cells
(incapable of autonomous replication)

All cells use nucleic acids (**RNA**) **to access stored information**

All cells use **proteins** as catalysts (enzymes) for chemical reactions

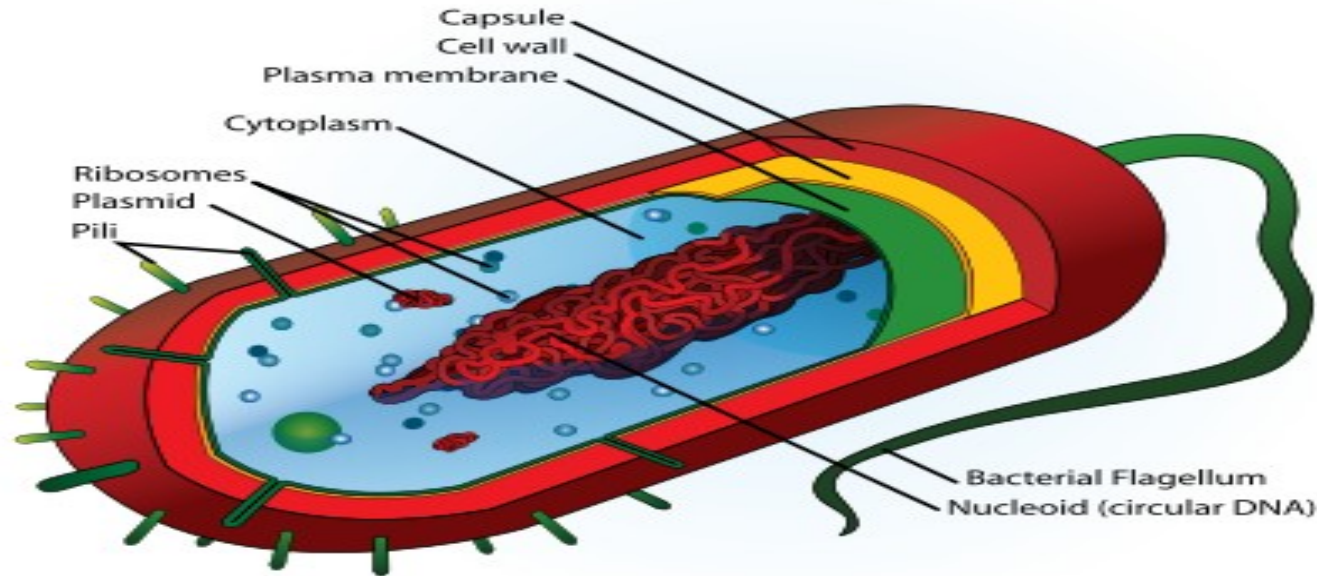
A few examples of RNA based enzymes, which may reflect primordial use of RNA

All cells use **lipids** for membrane components

Different types of lipids in different types of cells

All cells use **carbohydrates** for cell walls (if present), recognition, and energy generation

This is a prokaryotic cell



- Is a **single celled** organism that does **not** have a nucleus or membrane bound organelles.
- Its DNA is locked in the cytoplasm.

Two Types of Cells

Prokaryotic Cell Structure

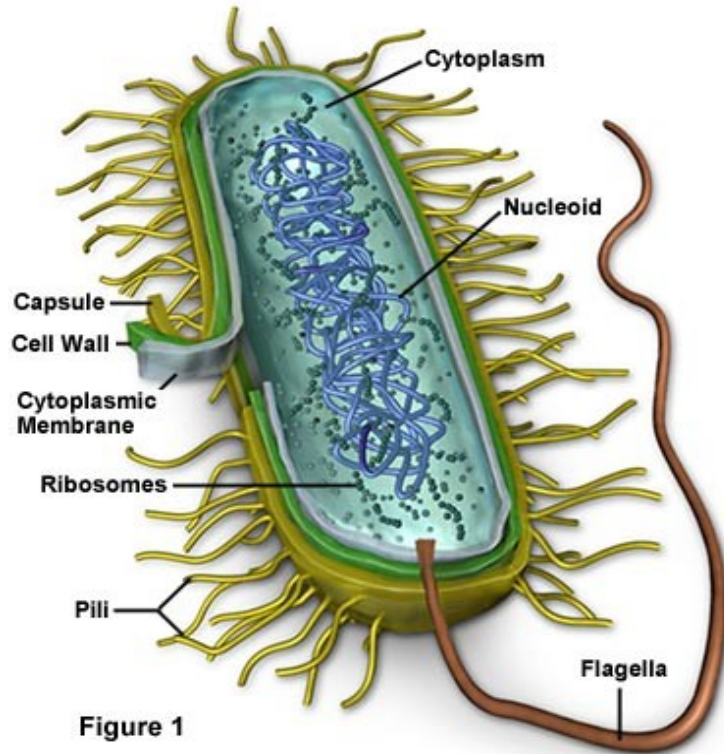
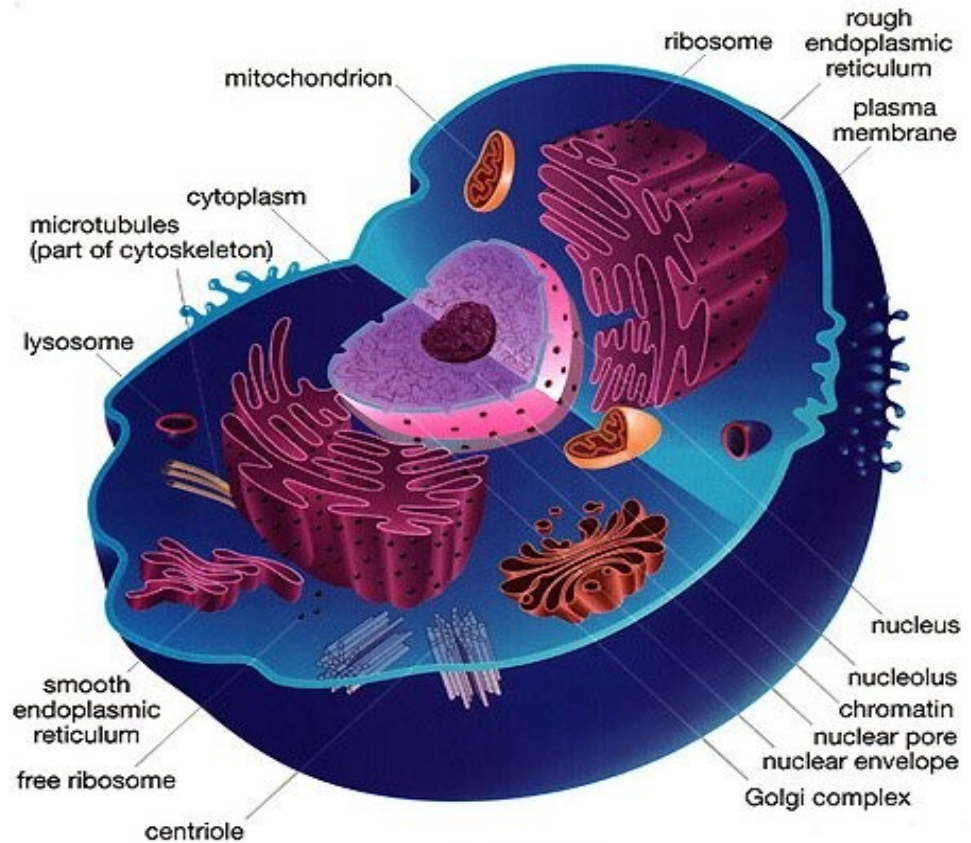


Figure 1

Nucleoid region contains the DNA

- Cell membrane & cell wall
- Contain ribosomes (no membrane) to make proteins in their cytoplasm



Contain 3 basic cell structures:

- Nucleus
- Cell Membrane
- Cytoplasm with organelles

The *E. coli* Cell and Human Cell

Prokaryotic cells

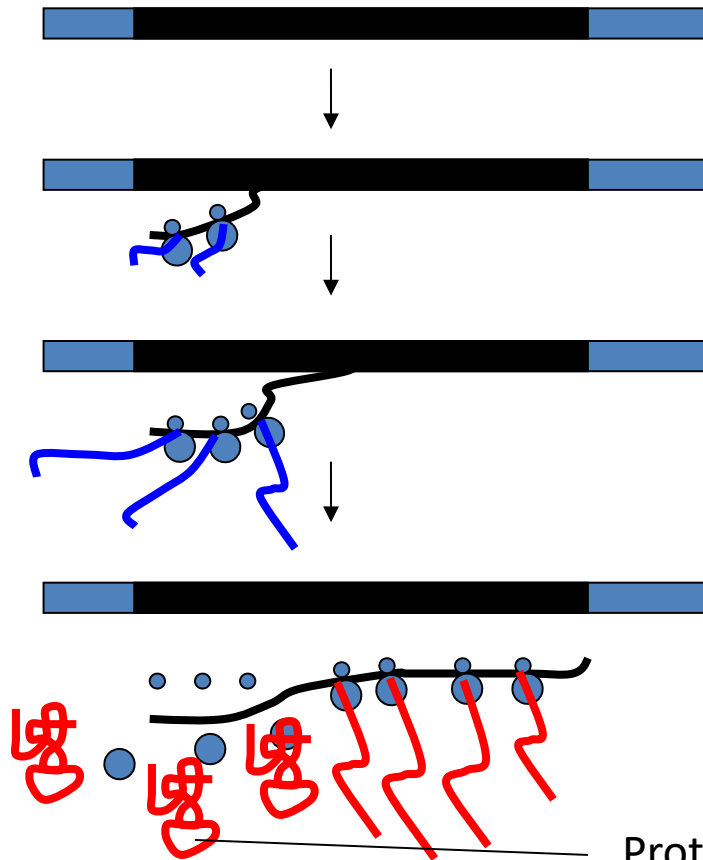
- They're **small and simple**, it grows and **divides fast**. It can adapt to rapid environmental changes.
- A **single compartment** with DNA and other biomolecules exist together in a well organized way.
- Extra-chromosomal DNA exists in the form of **plasmid**.

Eukaryotic cells

- A hundred times **larger**, much more **complex**
- **Contain organelles** to support different cellular functions from digestion to generation of energy and storage of nutrients.
- Extra-chromosomal DNAs exist in **mitochondria (animals) and chloroplast (plants)**.
- **Differentiate** to different cell-types.

Genetic information can be decoded by transcription

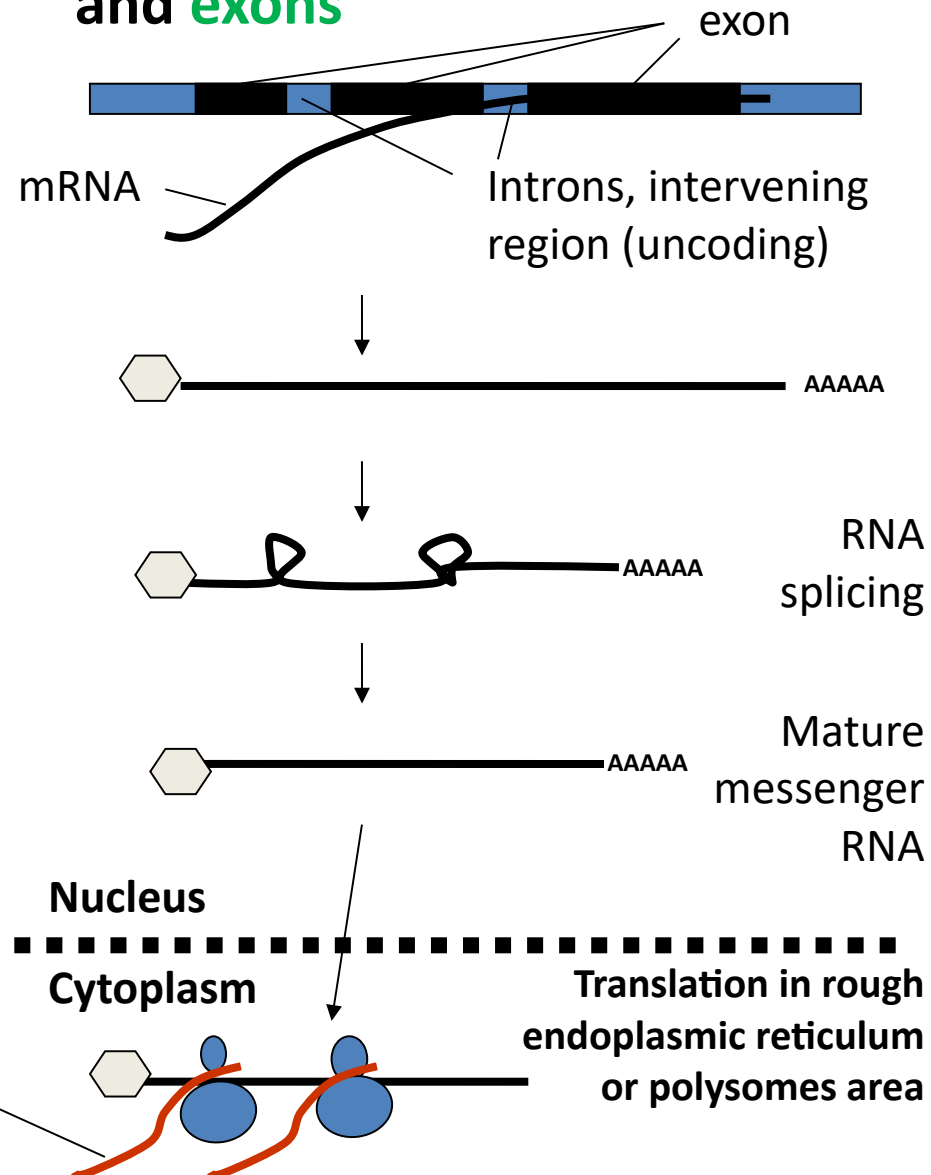
Prokaryotes: polycistronic, co-translation



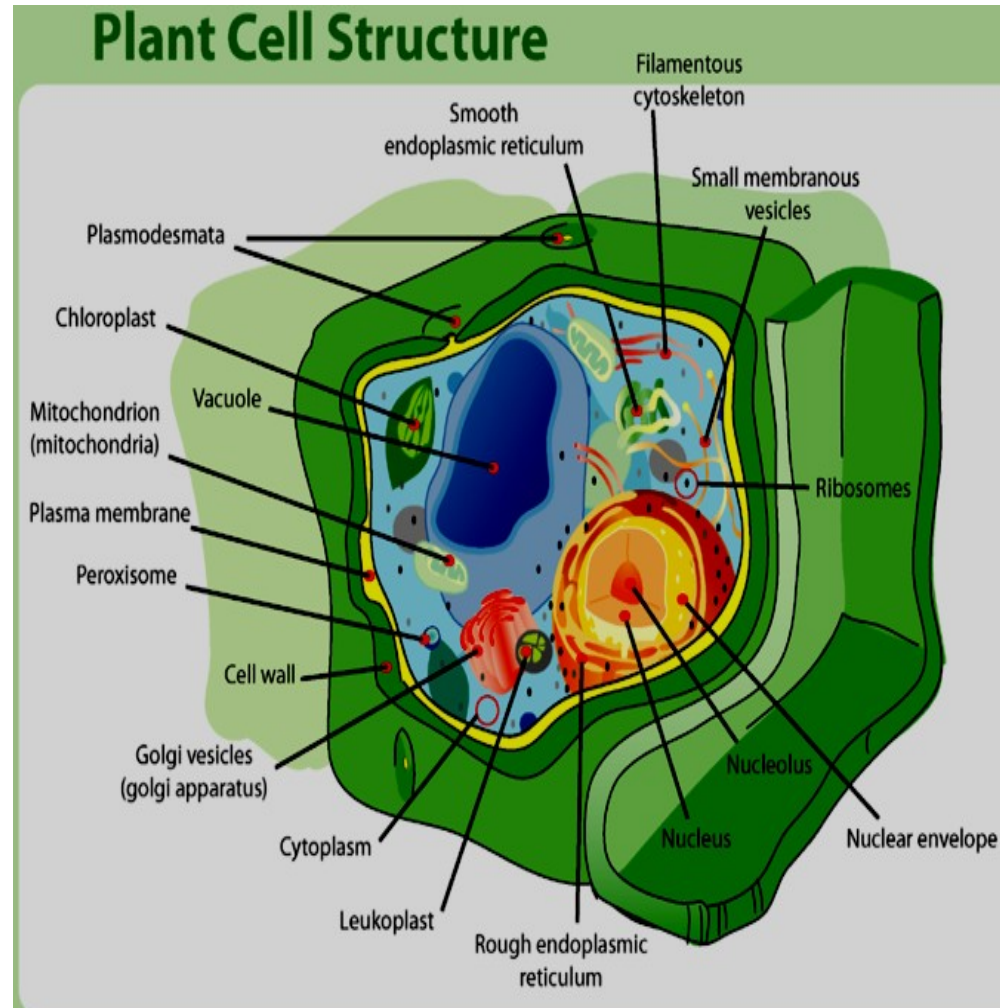
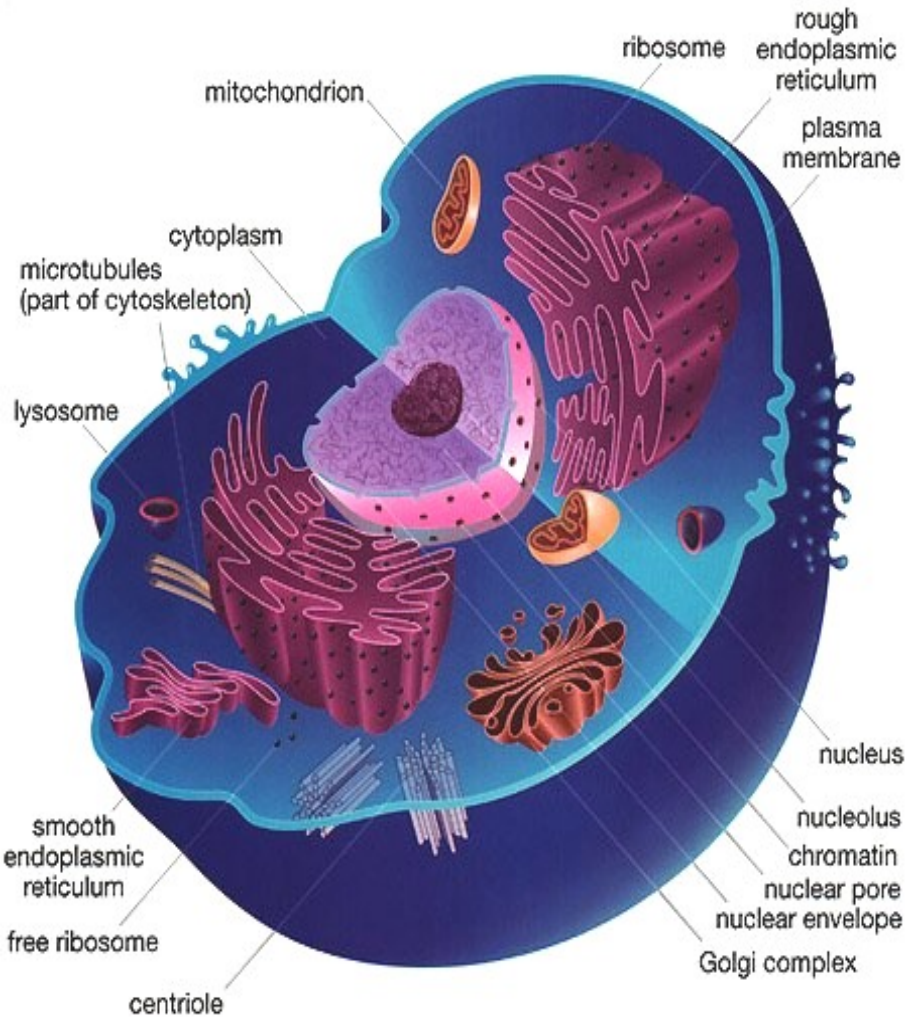
Polycistronic: one mRNA codes for several proteins.

Proteins made

Eukaryotes: genes have introns and exons



Two Main Types of Eukaryotic Cells



Characteristic Bio-membranes and Organelles

Plasma Membrane



- A lipid/protein/carbohydrate complex,
- Providing a **barrier** , **containing transport** and **signaling systems**.
- It holds the cell together and allows nutrients into the cell.

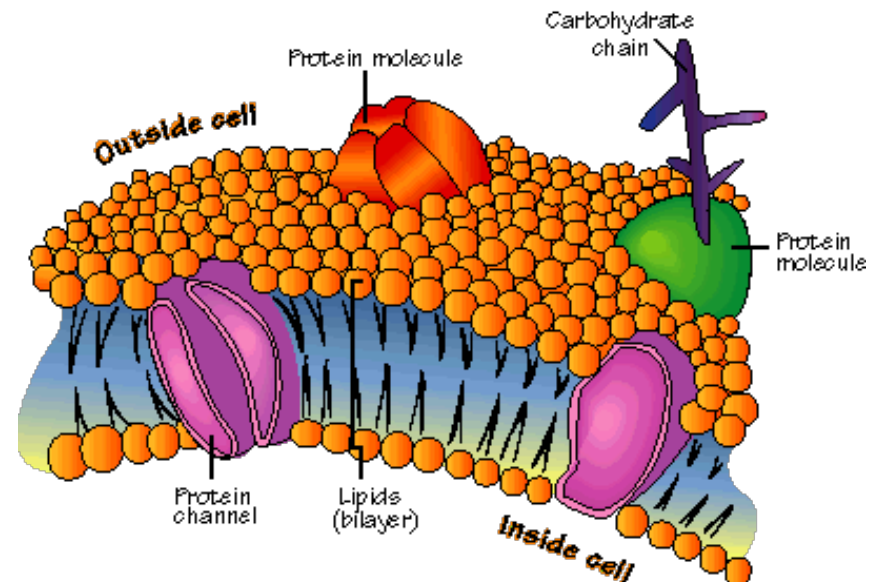
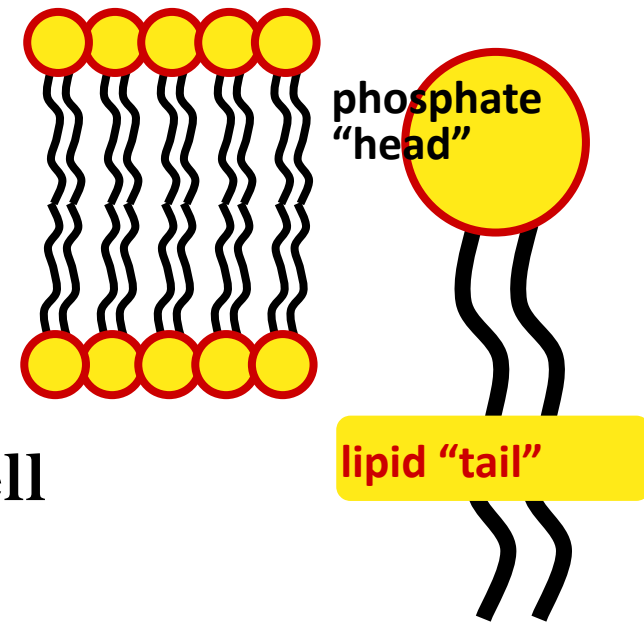
Cell membrane

■ Function

- ◆ separates cell from outside
- ◆ controls what enters or leaves cell
 - O_2 , CO_2 , food, H_2O , nutrients, waste
- ◆ recognizes signals from other cells
 - allows communication between cells

■ Structure

- ◆ double layer of fat
 - phospholipid bilayer
- ◆ receptor molecules
 - proteins that receive signals

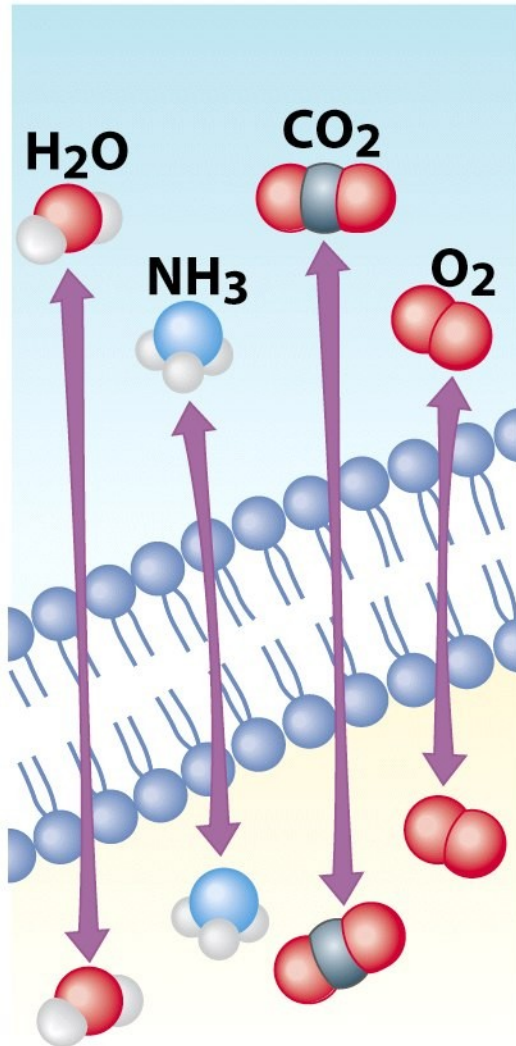


Movement Across the Plasma Membrane

- A few molecules **move freely**
 - Water, Carbon dioxide, Ammonia, Oxygen
- **Carrier proteins** transport some molecules
 - Proteins embedded in lipid bilayer
 - Fluid mosaic model – describes fluid nature of a lipid bilayer with proteins

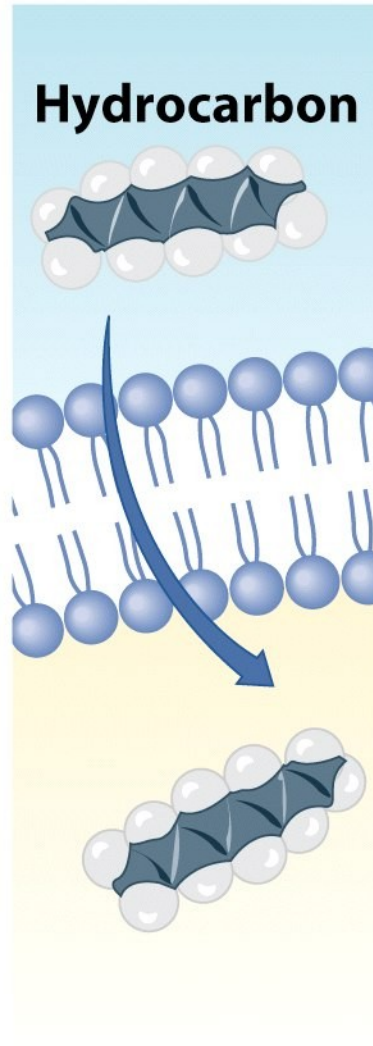
(a)

Small uncharged molecules



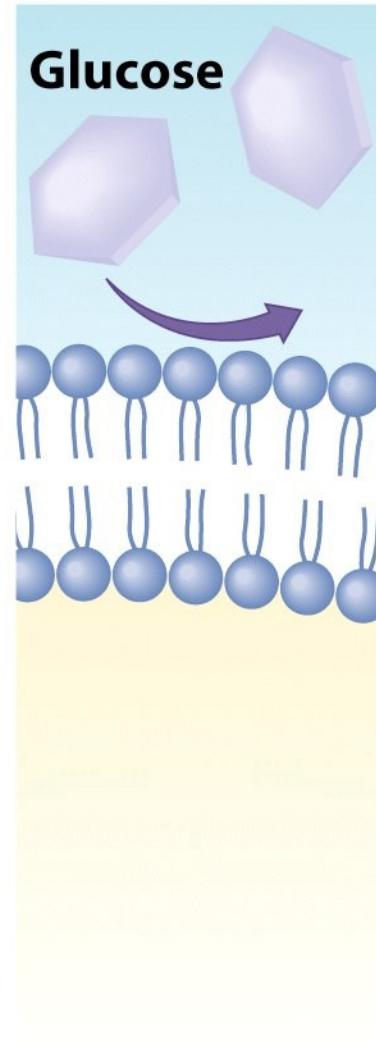
(b)

Lipid-soluble substances



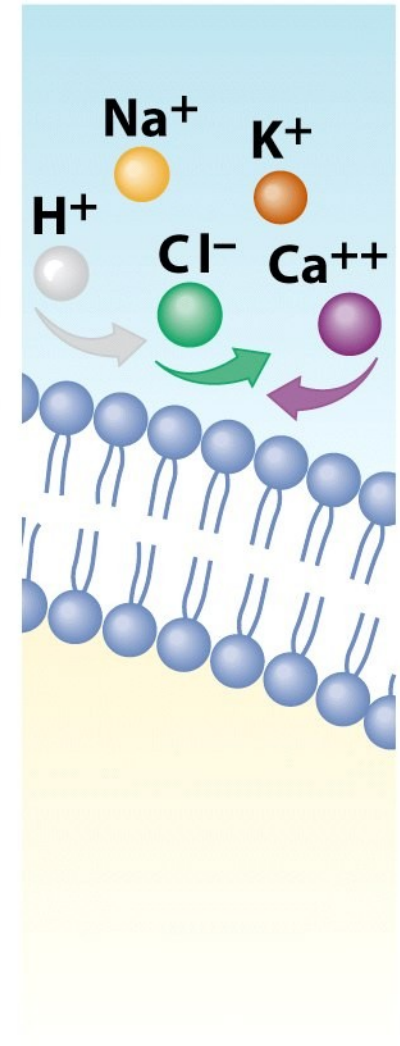
(c)

Water-soluble substances



(d)

Ions

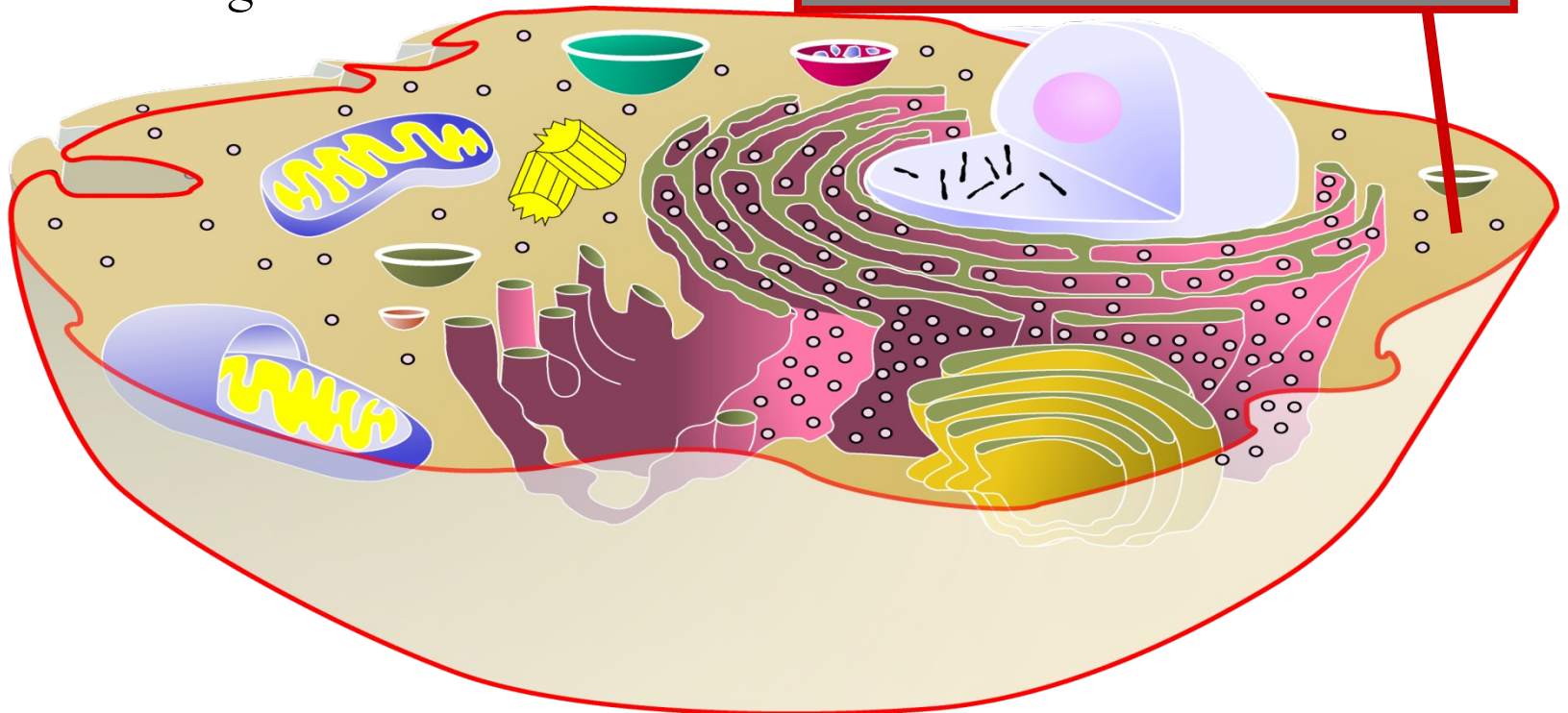


Cytoplasm

- Viscous fluid containing organelles
- components of cytoplasm
 - Interconnected filaments & fibers
 - Fluid = cytosol
 - Organelles (not nucleus)
 - storage substances

cytoplasm

▪ **jelly-like material holding organelles in place**



Nucleus

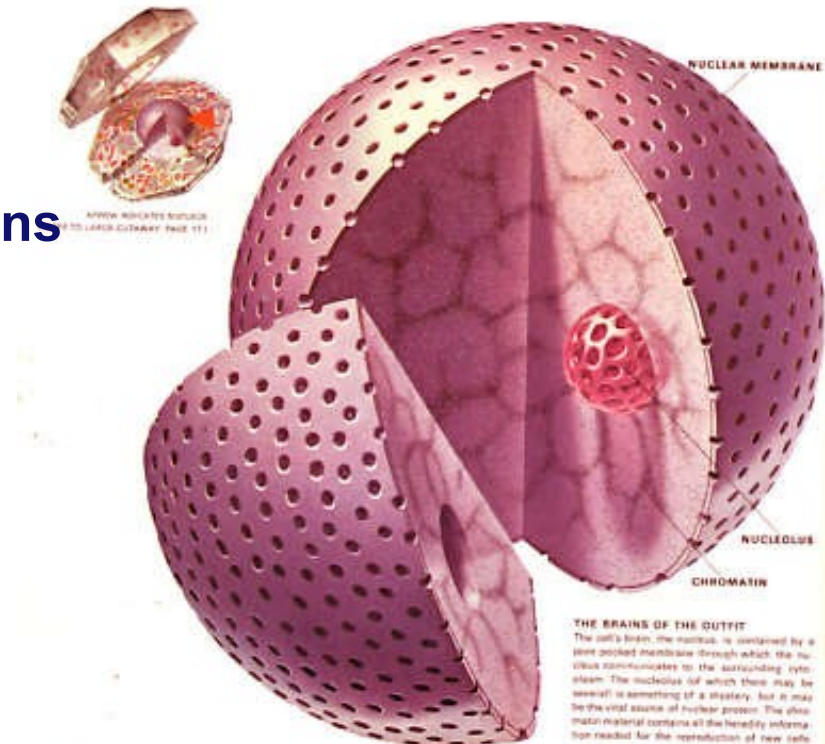
- Double membrane surrounding the chromosomes and the nucleolus.
- Pores allow specific communication with the cytoplasm.
- The **nucleolus** is a site for synthesis of rRNA making up the ribosome.

■ Function

- ◆ control center of cell
- ◆ protects DNA
 - instructions for building proteins

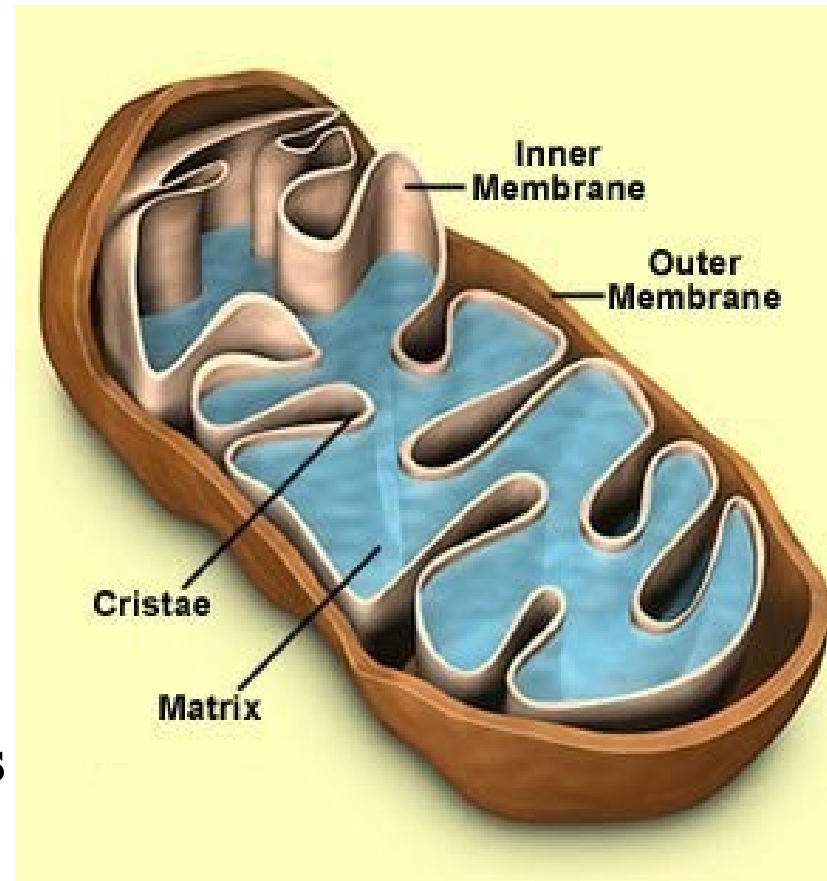
■ Structure

- ◆ nuclear membrane
- ◆ nucleolus
 - ribosome factory
- ◆ chromosomes
 - DNA



Mitochondrion

- Surrounded by a double membrane with a series of folds called **cristae**.
- Functions in **energy production** through metabolism.
- Contains **its own DNA**, and is believed to have originated as a captured bacterium



Features

- Mitochondria has got an **inner membrane** and an **outer membrane**. The space between these two is called **inter-membranous space**.
- Inner membrane convolutes into **cristae** and this **increases its surface area**.
- Both the membranes have different appearance and biochemical functions.

Biomedical importance

Inner membrane:

- It **surrounds** the matrix.
- It contains components of **electron transport system**.
- It is **impermeable** to most ions including H, Na, ATP, GTP, CTP etc and to large molecules.
- For the transport special carriers are present.
 - E.g. adenine nucleotide carrier(ATP –ADP transport).
 - Complex II- i.e. Succinate dehydrogenase .
 - Complex V- i.e. ATP synthase complex.

Outer membrane:

- It is **permeable** to most ions and molecules which can move from the cytosol to inter-membranous space.

Matrix:

- It is enclosed by the inner-mitochondrial membrane.
- Contains enzymes of citric acid cycle.
- Enzymes of β -oxidation of fatty acids.
- Enzymes of amino acids oxidation.
- Some enzymes of urea and heme synthesis.
- NAD, FAD
- ADP, Pi.
- Mitochondrial DNA.
- Mitochondrial cytochrome P450 system.

Mitochondrial cytochrome P450 system- it causes:

- a. Hydroxylation of cholesterol to steroid hormones (placenta, adrenal cortex, ovaries and testes)
- b. Bile acid synthesis (liver)
- c. Vitamin D formation(kidney).

Most **mitochondrial proteins** are derived from genes in **nuclear DNA**.

Mutation rate in mt DNA is 10 times more.

Mitochondrial Diseases:

Fatal infantile mitochondrial myopathy and renal dysfunction

MELAS(mitochondrial encephalopathy, lactic acidosis and stroke).

Lebers hereditary optic neuropathy

Myoclonic epilepsy

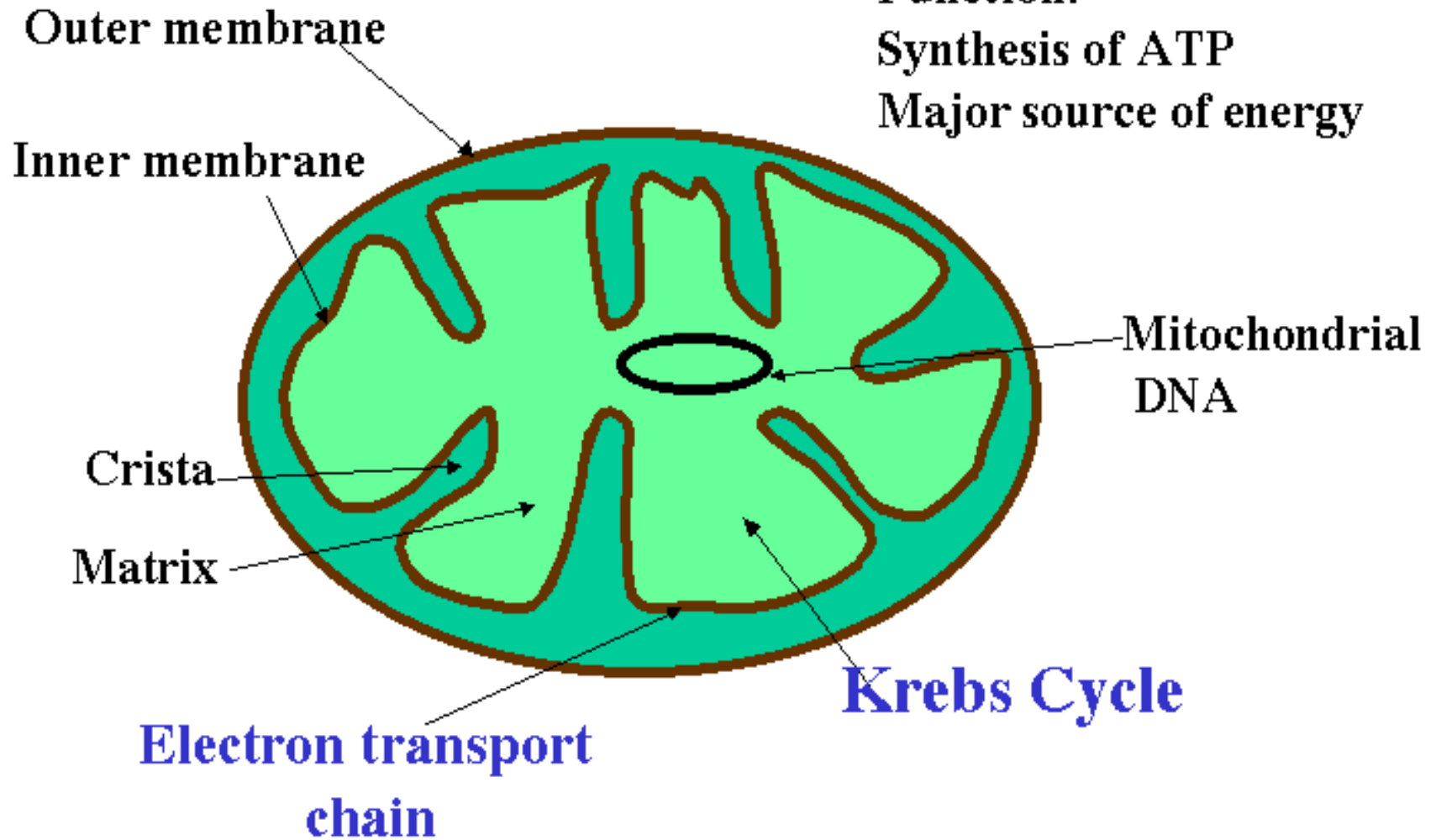
Ragged red fiber disease.

Also implicated in:

Alzheimer's disease, Parkinson's , Cardiomyopathies and diabetes.

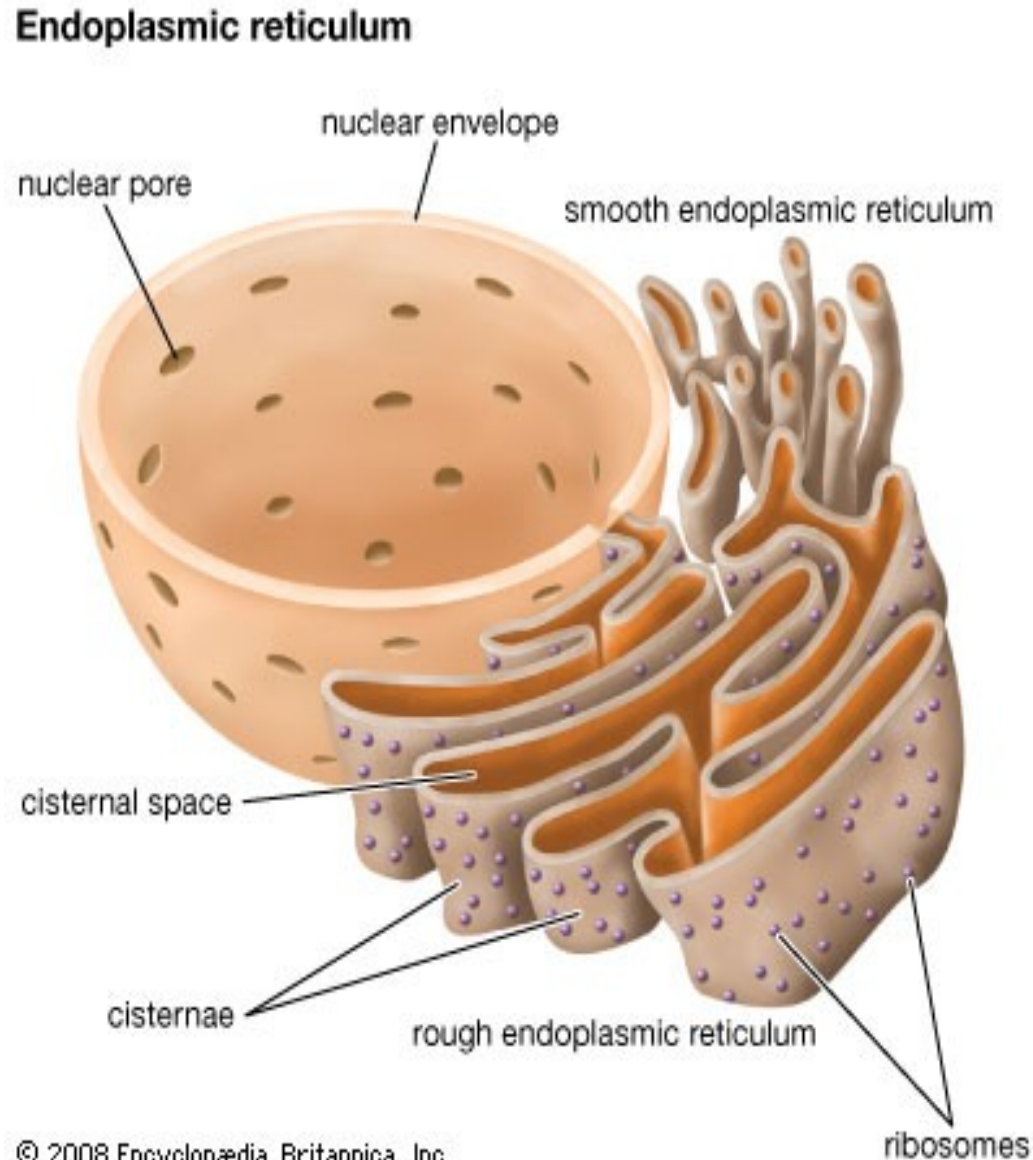
Mitochondria

Function:
Synthesis of ATP
Major source of energy



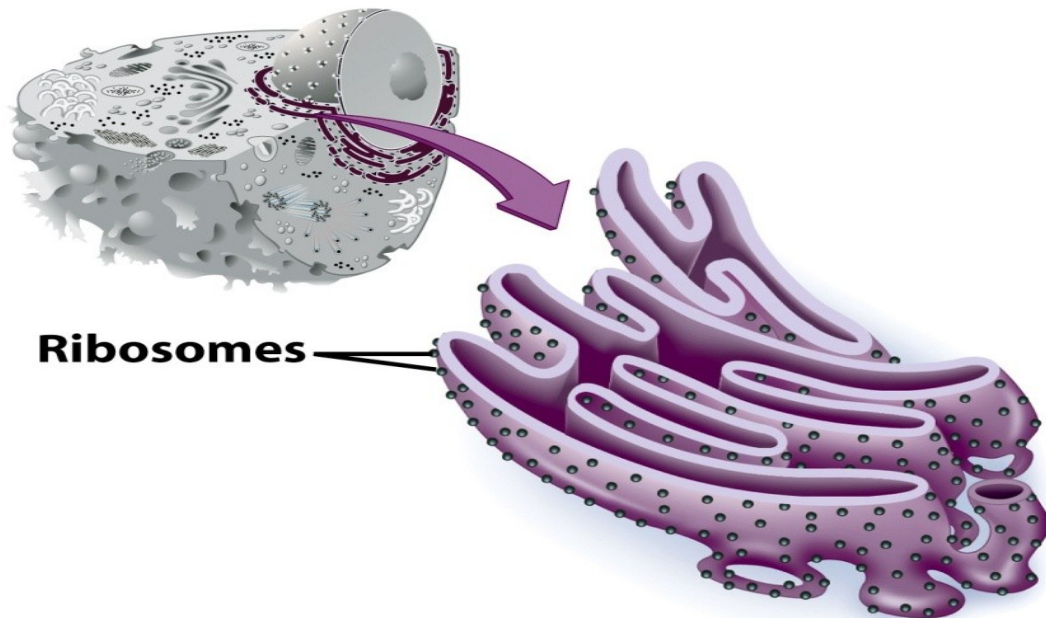
Endoplasmic Reticulum

- Helps move substances within cells
- Network of interconnected membranes
- Two types
 - Rough endoplasmic reticulum
 - Smooth endoplasmic reticulum



Rough Endoplasmic Reticulum(RER)

- Ribosomes attached to surface
 - Manufacture proteins
 - **Not all** ribosomes attached to rough ER
- May **modify proteins** from ribosomes

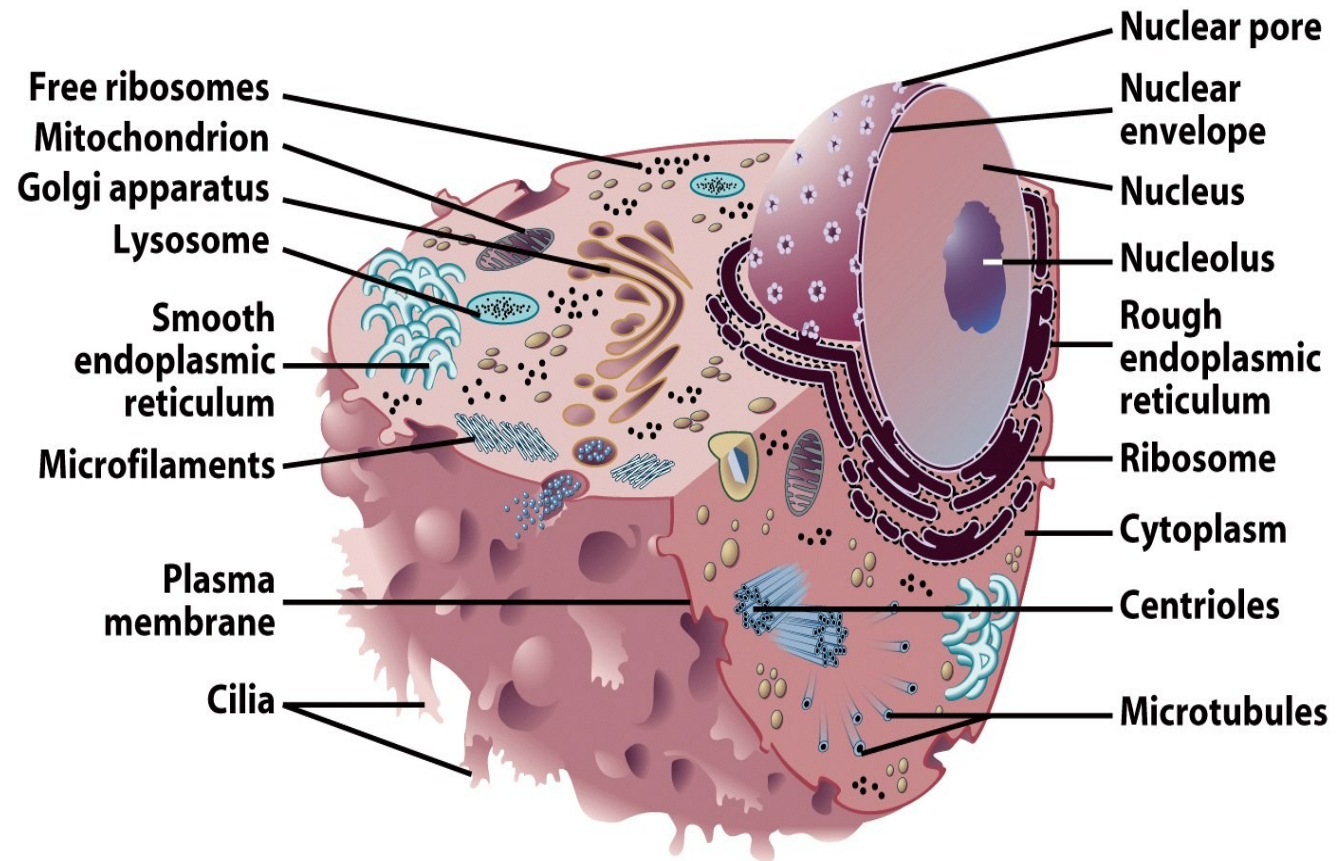


Biomedical importance of RER

- These membranes enclose a **lumen**.
- In this lumen newly synthesized proteins are **modified**.
- Rough appearance is due to the presence of **ribosomes** attached on its **cytosolic side(outer side)**.
- These ribosomes are involved in the **biosynthesis of proteins**, either incorporated into the **membranes or into the organelles**.
- Special proteins are present that are called **Chaperones**, play a role in proper folding of proteins.
- Protein **glycosylation** also occurs in ER. i.e. the carbohydrates are attached to the newly synthesized proteins.

Smooth Endoplasmic Reticulum

- **No** attached ribosomes
- Has enzymes that help build molecules
 - Carbohydrates
 - Lipids

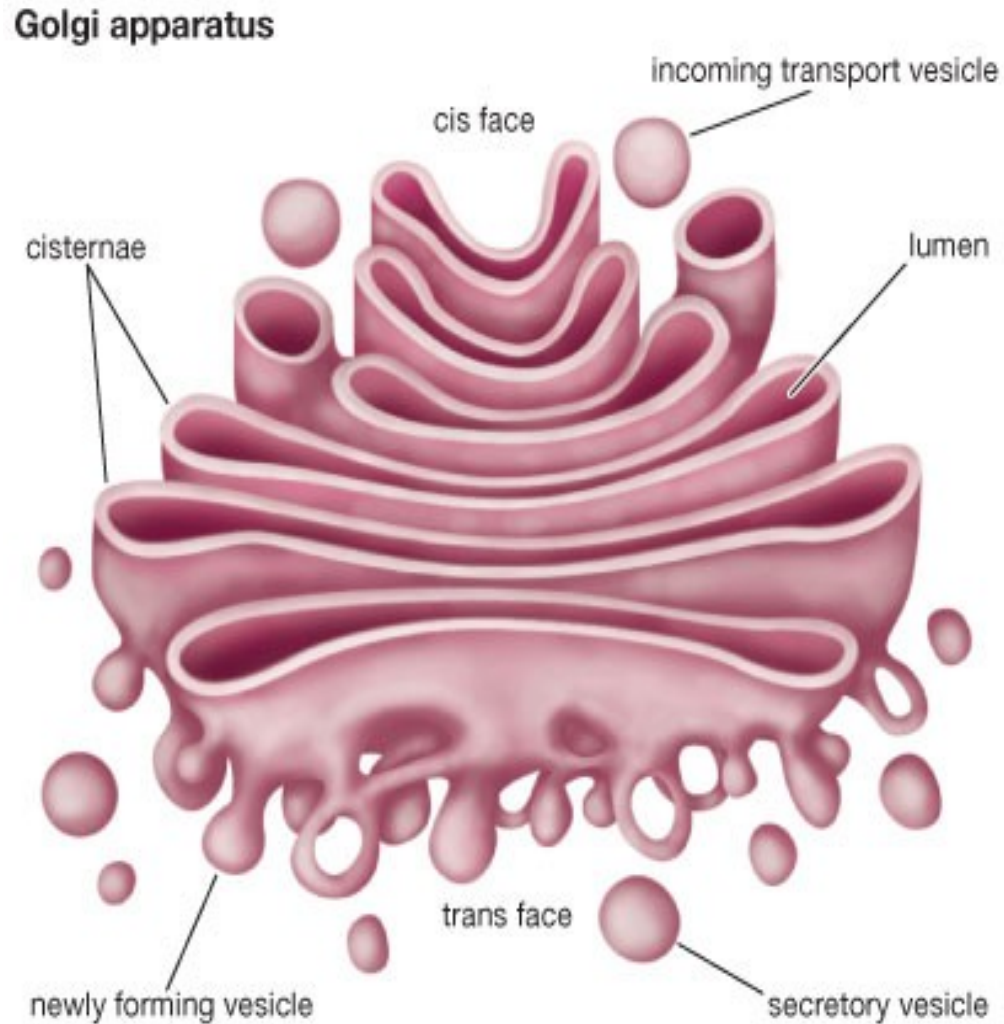


Smooth Endoplasmic Reticulum

- Involved **in lipid synthesis** like cholesterol synthesis and also in steroid hormones synthesis.
- **Detoxification** of endogenous and exogenous substances.
- The enzyme system involved in detoxification is called **Microsomal Cytochrome P450 monooxygenase** system(xenobiotic metabolism).
- ER along with Golgi apparatus is involved in the synthesis of other organelles –**lysosomes & Peroxisomes**.
- **Elongation** of fatty acids e.g. Palmitic acid 16 C- Stearic acid 18 C.
- **Desaturation** of fatty acids.
- **Omega oxidation** of fatty acids.

Golgi apparatus

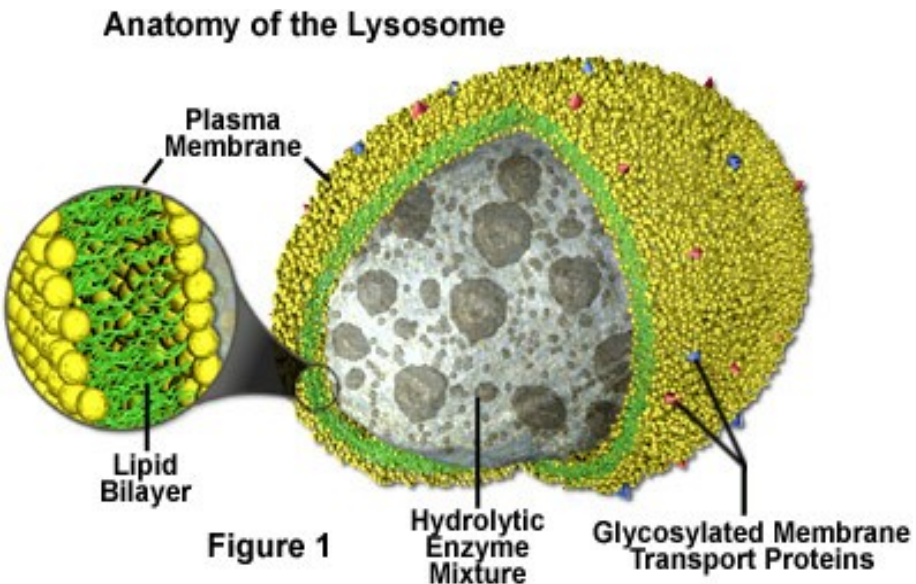
- Golgi complex is a network of flattened smooth membranous sacs- cisternae and vesicles.
- These are responsible for the **secretion of proteins** from the cells(hormones, plasma proteins, and digestive enzymes).
- It works in combination with ER.



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- Enzymes in golgi complex transfer carbohydrate units to proteins to form of **glycoproteins**, this determines the ultimate destination of proteins.
- Golgi is the major site for the synthesis of **new membrane, lysosomes and peroxisomes**.
- It plays two major roles in the membrane synthesis:
 - i. It is involved in the processing of oligosaccharide chains of the membranes (all parts of the GA participates).
 - ii. It is involved in the sorting of various proteins prior to their delivery(Trans Golgi network).

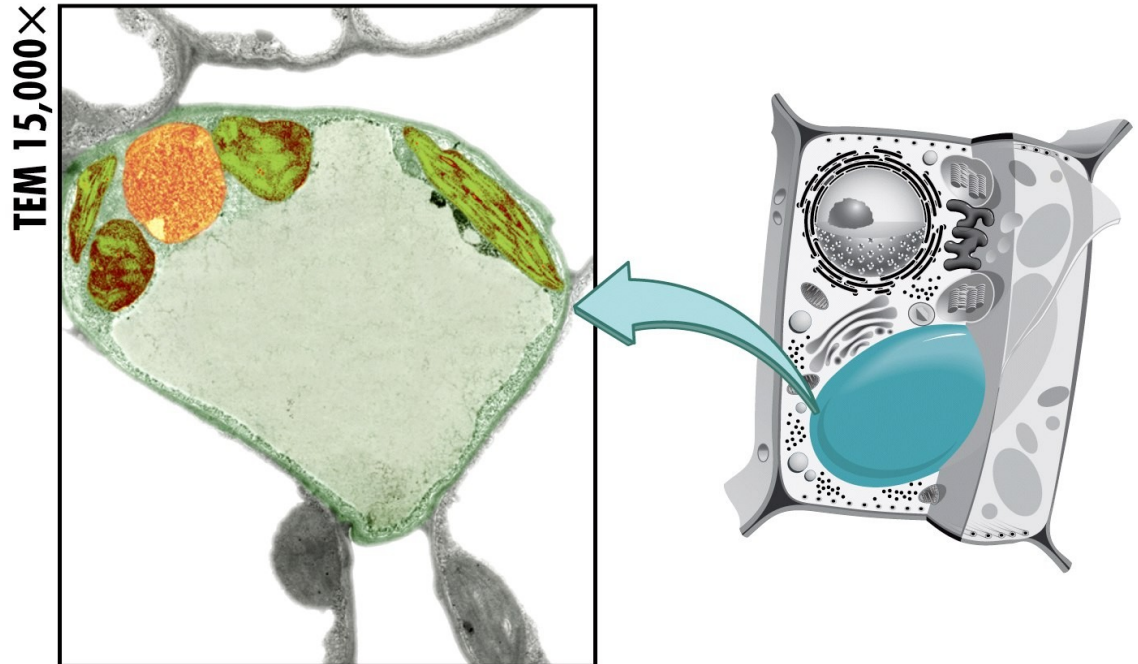
The Lysosome



- An intracellular digestive organ for both intra and extracellular substances.
- They have:
 - a single limiting membrane.
 - an acidic pH- 5
 - a group of enzymes called **Hydrolases**.
- **Acidic** condition to digest and remove unwanted materials or break down materials for re-use.

Vacuoles

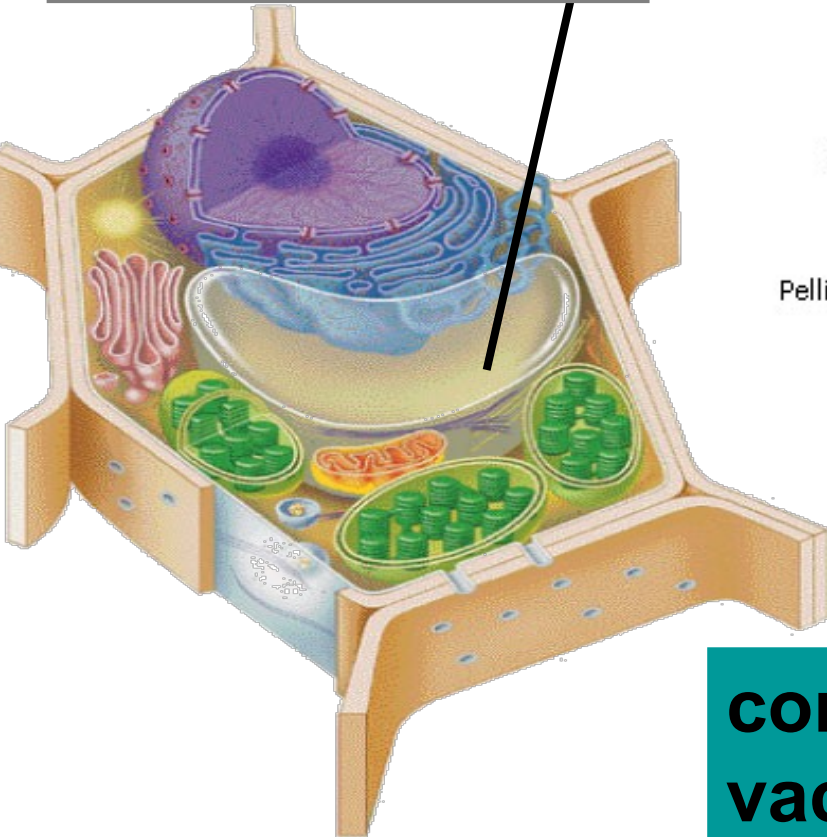
- Membrane bound storage sacs
- More common in plants than animals
- Contents
 - Water
 - Food
 - wastes



Food & water storage

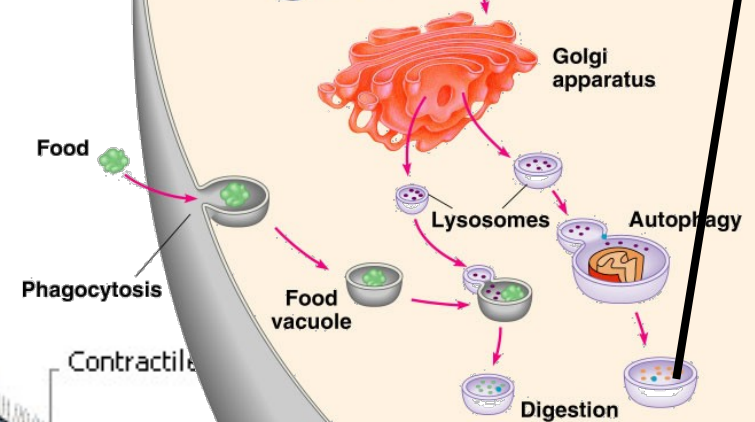
plant cells

central vacuole



Plasma membrane

food vacuole



Food vacuole

Contractile

Pellicle

Cilia

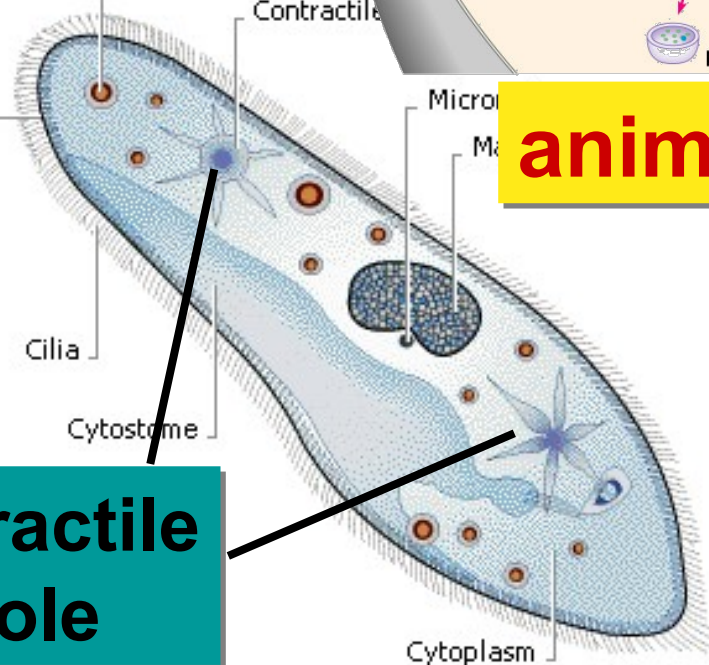
Cytostome

contractile vacuole

Micro

animal cells

Cytoplasm



Peroxisomes or Microbodies

- Called Peroxisomes because of their ability to produce or utilize H_2O_2 .
- They are small, **oval or spherical** in shape.
- They have a fine network of tubules in their matrix.
- About 50 enzymes have been identified.
- The **number of enzymes fluctuates** according to the function of the cells.

Biomedical importance

- Xenobiotics leads to the proliferation of Peroxisomes in the liver.
- Have an important role in the **breakdown of lipids**, particularly **long chain fatty acids**.
- **Synthesis** of glycerolipids, glycerol ether lipids, isoprenoids and bile.
- **Oxidation** of D- amino acids, Uric acid to allantoin (animals), Hydroxy acids which leads to the formation of H_2O_2 .
- Contain **catalase** enzyme, which causes the breakdown of H_2O_2 .

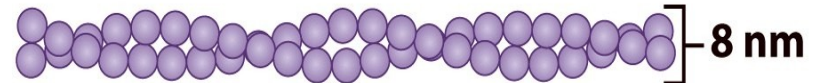
Diseases associated:

- Most important disease is **Zellweger Syndrome**.
- There is absence of functional peroxisomes.
- This leads to the accumulation of long chain fatty acids in the brain, decreased formation of plasmalogens, and defects of bile acid formation.

Cytoskeleton

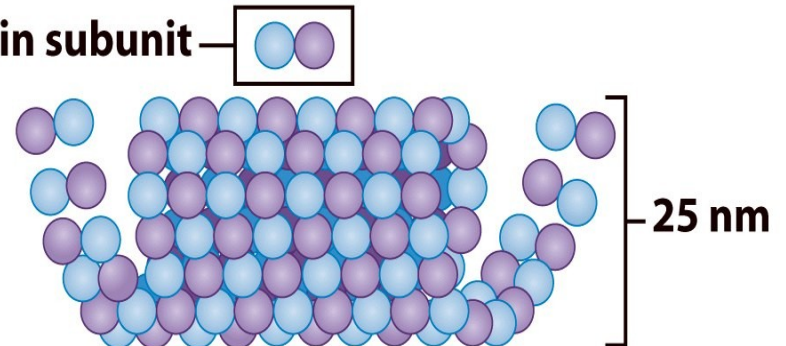
- Filaments & fibers
- Made of 3 fiber types
 - Microfilaments
 - Microtubules
 - Intermediate filaments
- 3 functions:
 - mechanical support
 - anchor organelles
 - help move substances

(a) Microfilament



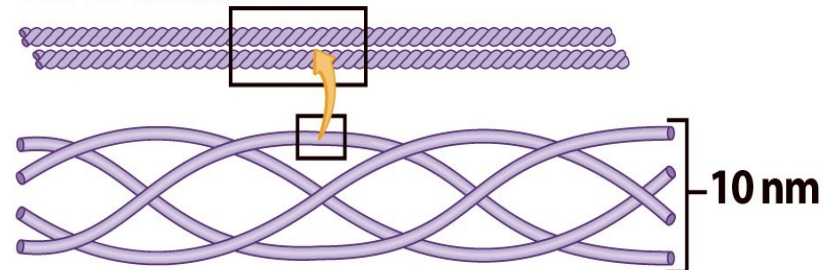
(b) Microtubule

Protein subunit



(c) Intermediate filament

Threadlike unit



1.3 Acid, Base and Buffer systems



Acid and Base Balance

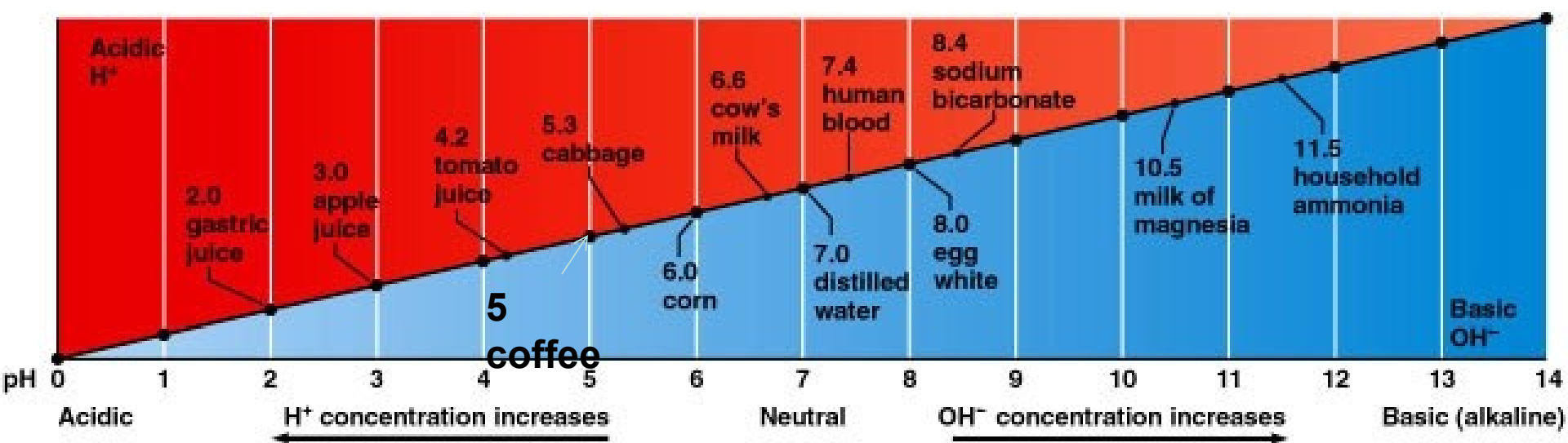
pH Review

$$\text{pH} = -\log [\text{H}^+]$$

- H^+ is really a proton

➤ Range is from 0 - 14

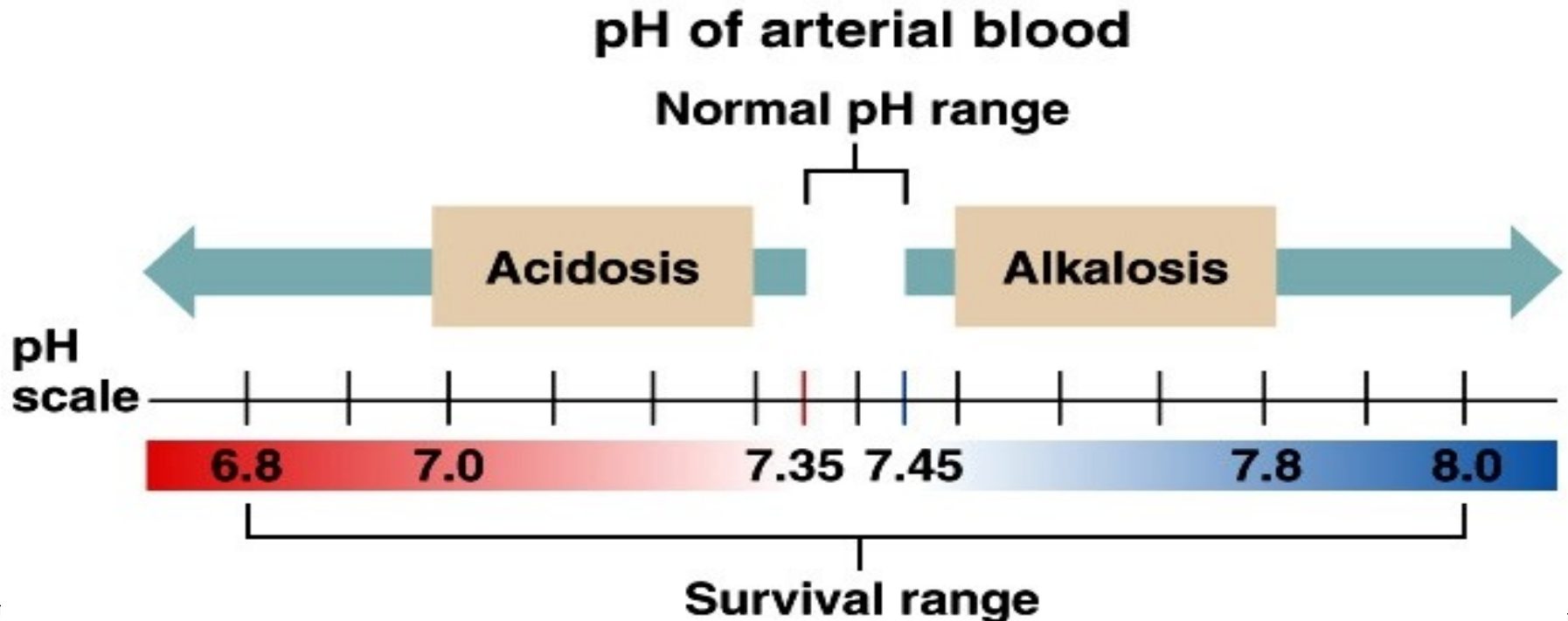
- If $[\text{H}^+]$ is high, the solution is acidic; $\text{pH} < 7$
- If $[\text{H}^+]$ is low, the solution is basic or alkaline ;
 $\text{pH} > 7$



Hydrogen Ion Concentrations and pH		
Grams of H^+ per Liter	pH	
0.0000000000000001	14	Increasingly basic
0.000000000000001	13	
0.00000000000001	12	
0.0000000000001	11	
0.000000000001	10	
0.0000000001	9	Neutral—neither acidic nor basic
0.000000001	8	
0.00000001	7	
0.0000001	6	
0.000001	5	
0.0001	4	Increasingly acidic
0.001	3	
0.01	2	
0.1	1	
1.0	0	

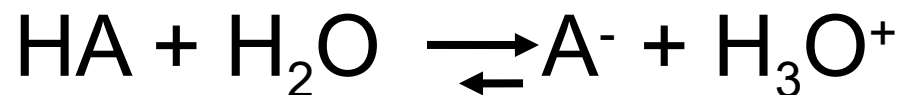
The Body and pH

- Homeostasis of pH is tightly controlled
- Extracellular fluid = 7.4
- Blood = 7.35 – 7.45
- ✓ If pH is < 6.8 or > 8.0 death occurs
- Acidosis (acidemia) below 7.35
- Alkalosis (alkalemia) above 7.45



- Acids are H^+ donors.
- Bases are H^+ acceptors.
- Acids and bases can be:
 - **Strong** – dissociate completely in solution
 - HCl , NaOH
 - **Weak** – dissociate only partially in solution
 - Lactic acid, carbonic acid
 - *A weak acid has a characteristic **dissociation constant**, K_a .*
- *The relationship between the pH of a solution, the K_a of an acid, and the extent of its dissociation are given by the **Henderson-Hasselbalch equation**.*

Acid/conjugate base pairs



HA = acid (donates H^+)(Bronstad Acid)

A^- = Conjugate base (accepts H^+)(Bronstad Base)

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$\text{p}K_a = -\log K_a$$

K_a & $\text{p}K_a$ value describe **tendency to loose H^+**

large K_a = stronger acid
small K_a = weaker acid

...cont'd

- *A **buffer** is a mixture of an **undissociated acid** and its **conjugate base** (the form of the acid having lost its proton).*
 - *It causes a solution to **resist changes in pH** when either H^+ or OH^- is added.*
- *A buffer has its **greatest buffering capacity** in the pH range near its **pK_a** (the negative log of its K_a).*
- *Two factors determine the effectiveness of a buffer, :*
 - ***its pK_a relative to the pH of the solution and***
 - ***its concentration.***

The Henderson–Hasselbalch Equation

- *Defines the relationship between pH and concentrations of conjugate acid and base.*
- A change in concentration of any component in an equilibrium reaction necessitates a concomitant change in every component.
- For example, an increase in $[H^+]$
 - will decrease the concentration of conjugate base (e.g., lactate ion)
 - with an equivalent increase in the conjugate acid (e.g., lactic acid).
- This relationship is conveniently expressed by
 - rearranging the equilibrium equation and
 - solving for H^+ , as shown for the following dissociation:

Henderson-Hasselbalch Equation

$$1) K_a = \frac{[H^+][A^-]}{[HA]}$$

HA = weak acid

A⁻ = Conjugate base

$$2) [H^+] = K_a \frac{[HA]}{[A^-]}$$

$$3) -\log[H^+] = -\log K_a -\log \frac{[HA]}{[A^-]}$$

$$4) -\log[H^+] = -\log K_a +\log \frac{[A^-]}{[HA]}$$

$$5) pH = pK_a +\log \frac{[A^-]}{[HA]}$$

* H-H equation describes the relationship between pH, pKa and buffer concentration

...cont'd

- Henderson and Hasselbalch equation is a convenient way of viewing the **relationship between**
 - **pH of a solution** and
 - **relative amounts of conjugate base and acid present.**
- When the ratio of [conj base]/[acid] is 1 : 1
 - pH equals the pK_a of the acid because $\log 1 = 0$, and thus $pH = pK_a$.
- If pH is **one unit less** than pK_a :
 - the [conj base]/[acid] ratio is **1 : 10**, and
- if pH is **one unit above** pK_a :
 - the [conj base]/[acid] ratio is **10 : 1**.

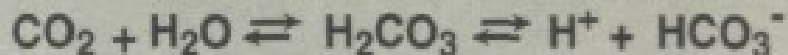
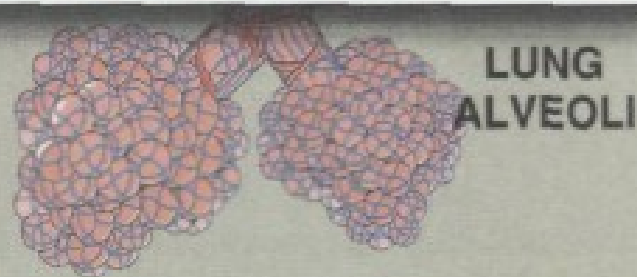
...cont'd

- If the pKa for a weak acid is known, this equation can be used
 - to calculate **the ratio of the unprotonated to the protonated form at any pH.**
- From this equation, you can see that **a weak acid is 50% dissociated at a pH equal to its pKa.**
- Most of the metabolic carboxylic acids have pKa 's between 2 and 5, depending on the other groups on the molecule.
 - **The pKa reflects the strength of an acid.**
- Acids with a pKa of 2 are stronger acids than those with a pKa of 5 because, at any pH, *a greater proportion is dissociated.*

Other applications of the Henderson-Hasselbalch equation

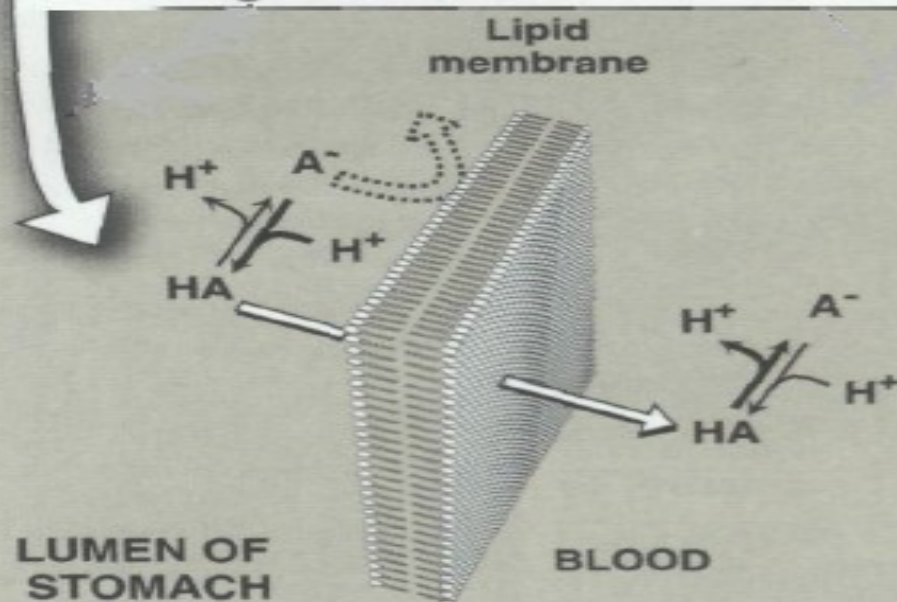
A BICARBONATE AS A BUFFER

- $\text{pH} = \text{pK} + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$
- An increase in bicarbonate ion causes the pH to rise.
- Pulmonary obstruction causes an increase in carbon dioxide and causes the pH to fall.



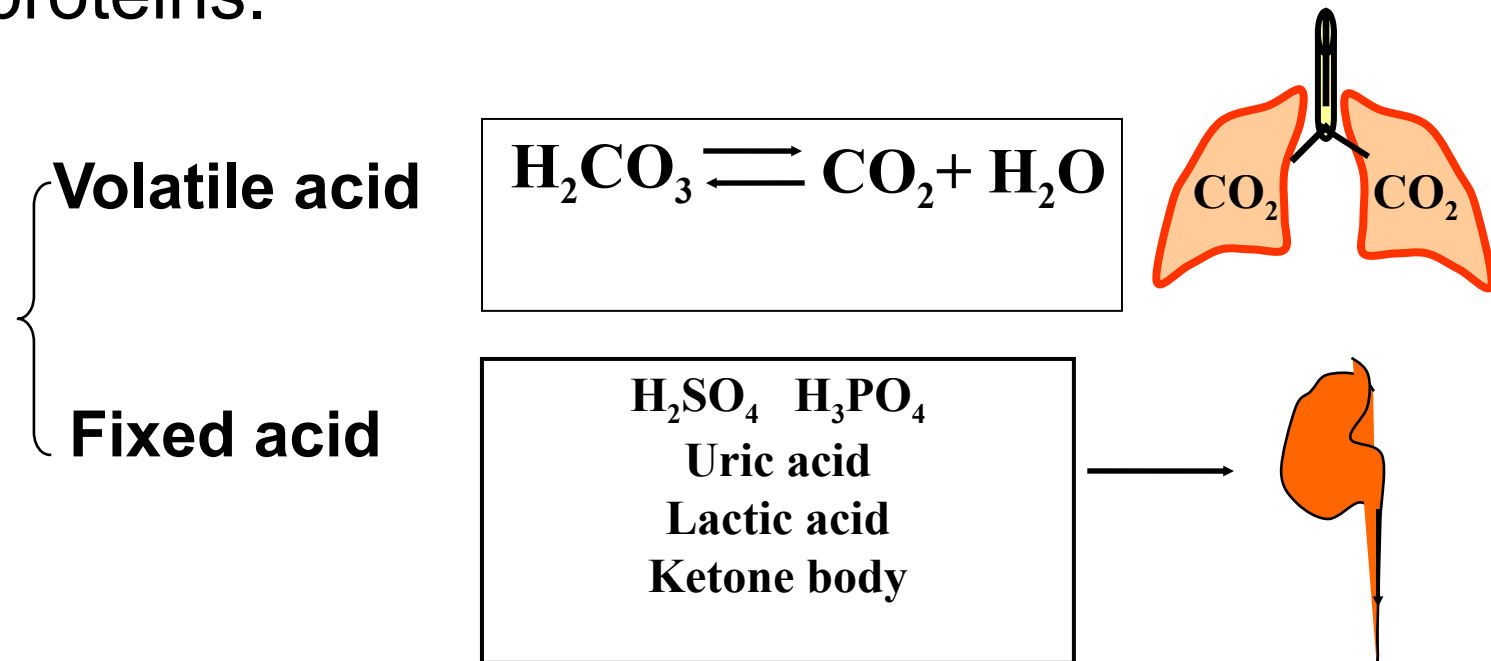
B DRUG ABSORPTION

- $\text{pH} = \text{pK} + \log \frac{[\text{Drug}^-]}{[\text{Drug-H}]}$
- At the pH of the stomach (1.5), a drug like aspirin (weak acid, $\text{pK} \approx 3.5$) will be largely protonated (COOH) and, thus, uncharged.
- Uncharged drugs generally cross membranes more rapidly than charged molecules.



The body produces more acids than bases

- Acids take in with foods.
- Cellular metabolism produces CO_2 .
- Acids produced by metabolism of lipids and proteins.



Maintenance of blood pH

- Three lines of defense to regulate the body's acid-base balance
 - Blood buffers
 - Respiratory mechanism
 - Renal mechanism

Buffer systems

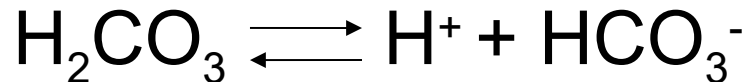
- Take up H^+ or release H^+ as conditions change
- Buffer pairs – weak acid and a base
- Exchange a strong acid or base for a weak one
- Results in a much smaller pH change

Principal buffers in blood

	in Plasma	in RBC
$\text{H}_2\text{CO}_3 / \text{HCO}_3^-$	35%	18%
HHb / Hb^-		35%
$\text{HPro} / \text{Pro}^-$	7%	
$\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$		5%
Total	42%	58%

Bicarbonate buffer

- **Predominant** buffer system in ECF
- Sodium Bicarbonate (NaHCO_3) and carbonic acid (H_2CO_3)
- HCO_3^- : H_2CO_3 : Maintain a **20:1** ratio



$$\text{pH} = \text{pK}_a + \lg \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

$$= 6.1 + \lg \frac{24}{1.2}$$

$$= 6.1 + \lg \frac{20}{1}$$

$$= 6.1 + 1.3 = 7.4$$

Bicarbonate buffer

- $\text{HCl} + \text{NaHCO}_3 \leftrightarrow \text{H}_2\text{CO}_3 + \text{NaCl}$
- $\text{NaOH} + \text{H}_2\text{CO}_3 \leftrightarrow \text{NaHCO}_3 + \text{H}_2\text{O}$

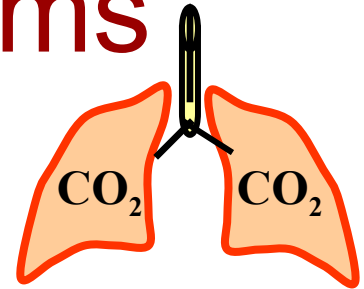
Phosphate buffer

- Major intracellular buffer
- $\text{NaH}_2\text{PO}_4\text{-Na}_2\text{HPO}_4$
- $\text{H}^+ + \text{HPO}_4^{2-} \leftrightarrow \text{H}_2\text{PO}_4^-$
- $\text{OH}^- + \text{H}_2\text{PO}_4^- \leftrightarrow \text{H}_2\text{O} + \text{HPO}_4^{2-}$

Protein Buffers

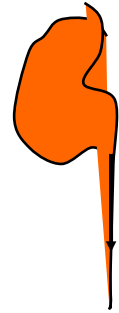
- Include plasma proteins and hemoglobin
- Carboxyl group gives up H^+
- Amino Group accepts H^+

2. Respiratory mechanisms



- Exhalation of CO_2
- **Rapid, powerful**, but only works with **volatile acids**
- $\text{H}^+ + \text{HCO}_3^- \leftrightarrow \text{H}_2\text{CO}_3 \leftrightarrow \text{CO}_2 + \text{H}_2\text{O}$
- **Doesn't** affect **fixed acids** like lactic acid
- Body pH can be adjusted by changing **rate and depth of breathing**

3. Kidney excretion



- Most effective regulator of pH
- The pH of urine is normally acidic (~ 6.0)
 - H^+ ions generated in the body are eliminated by acidified urine.
- Can eliminate large amounts of acid ($\rightarrow H^+$)
- Reabsorption of bicarbonate (HCO_3^-) ($\leftarrow HCO_3^-$)
- Excretion of ammonium ions (NH_4^+) ($\rightarrow NH_4^+$)
- If kidneys fail, pH balance fails

Rates of correction

- Buffers function: almost instantaneously
- Respiratory mechanisms: take several minutes to hours
- Renal mechanisms: may take several hours to days

**First line of
defense against
pH shift**

**Chemical
buffer system**

**Bicarbonate
buffer system**

**Phosphate
buffer system**

**Protein
buffer system**

**Second line of
defense against
pH shift**

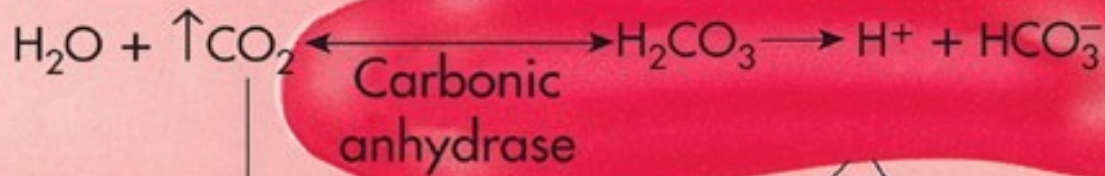
**Physiological
buffers**

**Respiratory
mechanism
(CO₂ excretion)**

**Renal
mechanism
(H⁺ excretion)**

Circulation

Erythrocyte



Lungs

Respiratory center in brain stem

$\uparrow$ Respiration rate and depth

$\uparrow\text{CO}_2$ given off

Kidney

$\downarrow\text{pH}$

$\uparrow$ Rate of H^+ secretion

Acid-Base Imbalances

- $\text{pH} < 7.35$: acidosis
- $\text{pH} > 7.45$: alkalosis
- The body response to acid-base imbalance is called **compensation**
 - The body gears up its homeostatic mechanism and makes every attempt to restore the pH to normal level.
 - May be **complete** if brought back within normal limits
 - **Partial compensation** if range is still outside norms.

Acid-Base Imbalances

- **Acidosis**- a decline in blood pH ↓
 - **Metabolic acidosis**: due to a decrease in bicarbonate. ↓
 - **Respiratory acidosis**: due to an increase in carbonic acid. ↑
- **Alkalosis**- a rise in blood pH ↑
 - **Metabolic alkalosis**: due to an increase in bicarbonate. ↑
 - **Respiratory alkalosis** : due to a decrease in carbonic acid. ↓

Compensation

- If underlying problem is **metabolic**, **hyperventilation or hypoventilation** can help: **respiratory compensation**.
- If problem is **respiratory**, **renal mechanisms** can bring about **metabolic compensation**.

Metabolic Acidosis

- **Bicarbonate deficit (↓)** - blood concentrations of bicarb drop below 22mEq/L (**milliequivalents / liter**)
- **Causes:**
 - **Loss of bicarbonate** through diarrhea or renal dysfunction
 - **Accumulation of acids** (lactic acid or ketones)
 - Failure of kidneys to excrete H^+
- Commonly seen in **severe uncontrolled DM (ketoacidosis)**.

Respiratory Acidosis

- **Carbonic acid excess** caused by blood levels of CO_2 above 45 mm Hg.
- **Hypercapnia** – high levels of CO_2 in blood
- **Causes:**
 - **Depression of respiratory center** in brain that controls breathing rate – drugs or head trauma
 - Paralysis of respiratory or chest muscles
 - Emphysema

Metabolic Alkalosis

- **Bicarbonate excess**↑ - concentration in blood is greater than 26 mEq/L
- Causes:
 - Excess vomiting = loss of stomach acid
 - Excessive use of alkaline drugs
 - Certain diuretics
 - Endocrine disorders: aldosterone ↑
 - Heavy ingestion of antacids

Respiratory Alkalosis

- Carbonic acid deficit↓
- $p\text{CO}_2$ less than 35 mm Hg (hypocapnea)
- Most common acid-base imbalance
- Primary cause is hyperventilation
 - Hysteria, hypoxia, raised intracranial pressure, excessive artificial ventilation and the action of certain drugs (salicylate) that stimulate respiratory centre.

Mixed acid-base disorders

- Sometimes, the patient may have two or more acid-base disturbances occurring simultaneously.
- In such instances, both HCO_3^- and H_2CO_3 are altered.

Summery

- Biochemistry is concerned with the **entire spectrum of life forms**, from relatively simple viruses and bacteria to complex human beings.
- Biochemistry, all life sciences and medicine are intimately related.
- **Health** depends on:
 - **a harmonious balance** of biochemical reactions occurring in the body, and
- **disease** reflects :
 - **abnormalities** in biomolecules, biochemical reactions, or biochemical processes.

- Advances in biochemical knowledge have illuminated many areas of medicine.
- Conversely, the study of diseases has often revealed previously unsuspected aspects of biochemistry.
- **Biochemical approaches** are often fundamental in illuminating the **causes of diseases** and **in designing appropriate therapies**.
- The judicious use of various biochemical laboratory tests is an integral component of **diagnosis** and **monitoring of treatment**.
- A sound **knowledge of biochemistry** and of other related basic disciplines is essential for the **rational practice of medicine and related health sciences**.

cytoplasm

- jelly-like material holding organelles in place

vacuole & vesicles

- transport inside cells
- storage

lysosome

- food digestion
- garbage disposal & recycling

nucleus

- protects DNA
- controls cell

mitochondria

- make ATP energy from sugar + O₂

cell membrane

- cell boundary
- controls movement of materials in & out
- recognizes signals

ER

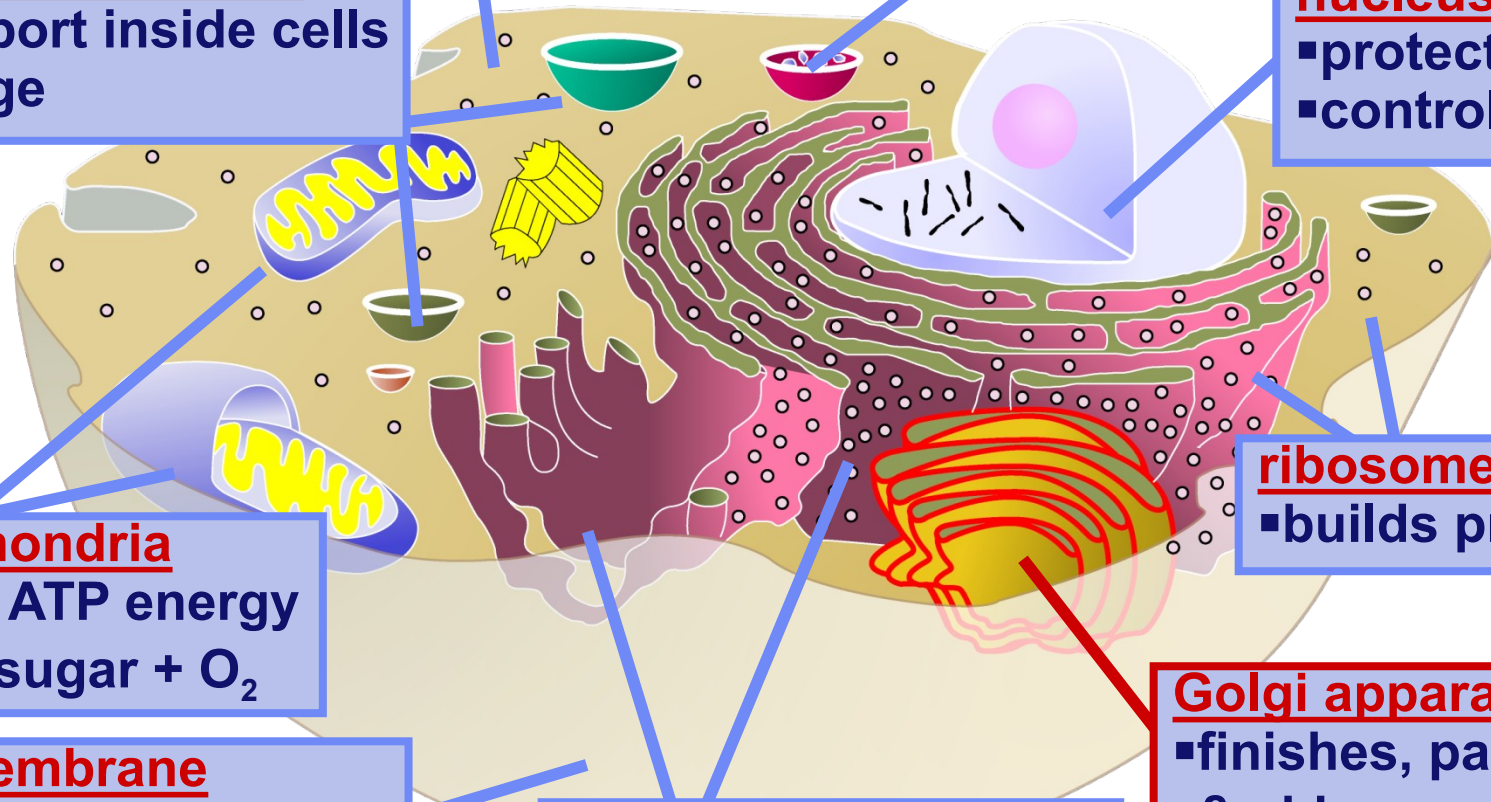
- helps finish proteins
- makes membranes

ribosomes

- builds proteins

Golgi apparatus

- finishes, packages & ships proteins



Summery

- Blood = 7.35 – 7.45;
- < 7.35: Acidosis, > 7.45: Alkalosis
- Three lines of defense to regulate the body's acid-base balance
 - **Blood buffers:** Bicarbonate buffer, Phosphate buffer, Protein Buffers
 - **Respiratory mechanisms:** Exhalation of CO₂
 - **Renal mechanism:** eliminate acid, Reabsorption of HCO₃⁻
- **Acidosis**- blood pH ↓ (Causes, Compensation)
 - Metabolic acidosis: bicarbonate ↓
 - Respiratory acidosis: carbonic acid ↑
- **Alkalosis**- blood pH ↑ (Causes, Compensation)
 - Metabolic alkalosis: bicarbonate ↑
 - Respiratory alkalosis : carbonic acid ↓

References

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Thank You!!